The Universal Right-Triangle Reflection Sequence (Unequal Legs)

A closed-form rate $r_n = 1 + 2\cos(2\pi/(n+3))$ and unconditional Class C faithfulness in every dimension

Vico Bonfioli*

June 15, 2026

Abstract

For every right triangle T in the plane with positive **unequal** rational legs, the BFS orbit-growth sequence u_d^T under the group of side-reflections is T-independent through depth 32, and this is sharp: the $(1,2)$ triangle deviates at depth 33 (Theorem 11); we call the common sequence the *universal right-triangle reflection sequence*. The natural conjecture that this independence persists at all depths is **refuted**: for every rational — indeed every algebraic — shape we construct an explicit nontrivial word in $\ker \rho_T$ (a product of conjugated glide-reflection squares realizing the translation $2(t^{-1}-1)\mu_T(t)$, which evaluates to 0 at the rotation number ζ_T), so each fixed shape deviates from the universal sequence at a first depth n_T sandwiched by $\max(31, c_T/2) \le n_T \le 24c_T + 8$ (Theorem 15, Corollary 16). The underlying structure theorem computes the translation lattice of the generic group exactly — $\mathcal{T} = 2(t-1)\mathbb{Z}[t^{\pm 1}]$, by a mod-2 collapse of the two legs onto a single mirror that degenerates the group to the infinite dihedral group (Theorem 13) — whence every algebraic specialization kernel is classified: $\ker \rho_T \cong \mathbb{Z}[t^{\pm 1}]$, normally generated by the single witness w_T (Corollary 14). The universal sequence is thus exactly the orbit growth of the *generic* right triangle, realized precisely by transcendental leg ratios (Corollary 19). The translation kernel of the side-reflection representation is computed via Mayer–Vietoris and Crowell as the $\mathbb{Z}[Q]$-module $I_A \cap I_B$, and this is *proved* to be the cyclic ideal $\mathbb{Z}[Q] \cdot h$ on the central generator $h = (c-1)(Y-1)$, via the exactness of the Bass–Serre tree of D_∞ (Lemma 29); consequently the generic kernel is identified abstractly: $\ker \rho_{\text{gen}} = [T_\rho, T_\rho]$ — the generic group is the Q-relative metabelianization of $V_4 * \mathbb{Z}/2$ (Corollary 31). In every dimension $n \ge 2$, the right-corner orthoscheme reflection group has closed-form Coxeter–Steinberg growth rate $r_n = 1 + 2\cos(2\pi/(n+3))$ (Theorem 38). For pairwise distinct legs the affine representation is unconditionally faithful in every dimension $n \ge 3$ (Theorem 42), via a uniform Schur-complement determinant identity $\det Q_n = -\prod_i a_i^2$ (Lemma 41) together with rank-0 descent on the three Cremona curves `24a1`, `32a2`, and `72a2`. A dimension-independent collision-depth invariant $\mathrm{cd}_n(\lambda) \in \{\infty, 4, 3\}$ determined by the (e,t)-partition statistics of the leg sequence (Theorem 43) classifies first-collision behaviour across all dimensions. A triangle-independent symbolic normal form reduces extra collisions to the vanishing of bounded-height integer Laurent polynomials at the rotation number $\zeta_T = (a+bi)/(a-bi)$; Gauss's lemma applied to the shape-dependent minimal polynomial μ_T then yields an effective all-depths theorem — a deviation at depth d forces c_T to divide both extreme coefficients of a realized word-difference, so every triangle with $c_T > 2d$ is universal at depth d (Theorem 10) — and, combined with a finite modular certificate over the sparse remaining candidates, sharp uniform universality through depth 32 (Theorem 11). The eight length-10 relations, the Schur-complement identity, and the T-independence of the orbit-growth sequence through depth 22 are formally machine-verified in the Lean 4 proof assistant (§7). For the 2D universal sequence the lamplighter transfer yields an explicit q-difference functional equation, and unfolding it pins the asymptotic ratio to the root of an explicit singularity equation: $\beta_2 = 1.49161778\ldots$ *exactly* (Proposition 26), the true value, since the travel block carries

*Independent researcher, Italy. `vicobonfioli@gmail.com`. https://elvec1o.github.io/home/posts/.

the dominant singularity and is connectivity-invariant (every pure-travel run is a single Euler trail), squeezing the true rate between its pure-travel subcount and the relaxed series; this *excludes* the value 3/2 that the finite-depth ratios had suggested. The closed form also yields transcendence of the relaxed generating function $V(x)$ (Theorem 27), modulo one explicit analytic lemma — the phase-uniform cosine asymptotic $S_1 = 1 - \cos\sqrt{2/(-\ln q)} + O(\sqrt{-\ln q})$, which we reduce by a monotone-domination Abel identity to a single oscillatory-average bound $\sum_j \mu_j R_j(w) = O(\sqrt{\tau})$ (probability weights μ_j, cosine Taylor remainders R_j; verified to 110 digits, bulk-block leading term in closed form $-\frac{17\sqrt{2}}{36}\sqrt{\tau}\sin w$, dominant travel block $+\frac{\sqrt{2}}{36}$). Its hard oscillatory core is resolved by model subtraction: subtracting the closed-form geometric kernel and expanding the residual through the exact resummation engine $\sum_j j^p e^{-j\tau}(-1)^j u^{2j}/(2j)! = (-\partial_\tau)^p \cos(ue^{-\tau/2}) = D_p$ yields closed-form layers; L_1 is rigorous, and the tail cancellation is captured exactly by the Touchard/Poisson identity $D_p(W) = 2^{-p}\mathrm{Re}[e^{iW}T_p(iW)]$, $T_p(w) = \mathbb{E}[N^p]$ ($N \sim \mathrm{Poisson}(w)$), which with the proved R-control $0 \le R_j \le \tau^2 C(j)$ and a finite truncation of the τ-series (the full absolute majorant diverges in the Poisson tail) gives $\sum_{m\ge 2} \sup|L_m| = O(\tau)$ and hence $\sup|\mathrm{corr}| \le 0.20\tau$, i.e. the lemma. This reduces transcendence of V to verification of an explicit elementary (if intricate) estimate with no remaining divergent-majorant gap; pending that refereeing we state V transcendental modulo the lemma rather than claim it unconditionally here. The constituent blocks $\Sigma_1, \Sigma_0, S_1, S_0$ are however transcendental *unconditionally*: each is an integer power series of radius 1 and non-rational, so by Pólya–Carlson has the unit circle as a natural boundary (this does not transfer to V, whose boundary singularities cancel in the ratio). The true series U shares V's travel denominator $1 - \Sigma_1$ verbatim, so its transcendence reduces, conditionally on the same lemma, to a single numerator non-cancellation at the infinitely many poles accumulating at $q = 1$; it remains conjectural (Conjecture 28).

1. Setting

Let T be a triangle with vertices $V_0, V_1, V_2 \in \mathbb{Q}^2$. Only the vertex coordinates and the leg lengths a, b are assumed rational. For $i \in \{0,1,2\}$ write R_i for the reflection of $\mathbb{R}^2$ across the line containing the side opposite V_i. Set $W(T) = \langle R_0, R_1, R_2\rangle \le \mathrm{Isom}(\mathbb{R}^2)$.

When we need explicit coordinates (from §3 onward) we place the right-angle vertex at $B = (a, 0)$ with $A = (0, 0)$, $C = (a, b)$, and write a, b for the two leg lengths.

We explore the orbit of T under $W(T)$ by breadth-first search in the Cayley graph relative to the generating set $\{R_0, R_1, R_2\}$. Let

$$\mathcal{O}_d(T) = \{\, g \cdot T : g \in W(T) \text{ has minimal word-length } d \,\},$$

and set $u_d := |\mathcal{O}_d(T)|$, $u_0 = 1$. All distances are computed in exact rational arithmetic and the proofs are checked symbolically over $\mathbb{Q}(a, b)$.

1.1 Notation

- u_d: BFS layer size at minimal word-length d.
- $\delta_d := F_{d+3} - u_d$: Fibonacci deficit against the Steinberg/Fibonacci baseline.
- $\rho_T : W \to \mathrm{Aff}(\mathbb{R}^2)$: the affine action of $W = \langle R_0, R_1, R_2\rangle$ induced by the sides of T.
- $\mathrm{cd}_n(\lambda)$: the collision-depth invariant in dimension n for a leg sequence λ (Theorem 43).
- (e, t): positional invariant of a leg sequence; e counts endpoint-adjacent equal pairs and t counts triples of consecutive equal legs.

Related work. The n-dimensional rate $r_n = 1 + 2\cos(2\pi/(n+3))$ derived in §6 is the closed form expected for the growth rate of a path-shaped Coxeter group with one ∞ edge, in the framework of Cannon–Floyd–Parry (Cannon–Wagreich; Floyd–Plotnick). The same family of algebraic integers also appears in birational surface dynamics (Diller–Favre; McMullen; Bedford–Kim); the overlap is at the level of expressions, not of number-theoretic class. Isola–Piergallini (*On the generic triangle group*, 2014–2015) establish that the reflection group of a generic triangle is the expected free product / Coxeter group with no unexpected relations; our unequal-leg right-triangle results are the precedent's exact rational analogue, replacing genericity by an explicit Diophantine ($\mathbb{Q}$-rational) faithfulness criterion and adding the closed-form growth data. The rotation number $\zeta_T = (a+bi)/(a-bi)$ governing our translation module is exactly the rotation number of the generalized-pinwheel *substitution* tilings of Sadun, built from a right triangle, and of the statistical circular symmetry studied by Frettlöh; those works analyze tiling-space invariants (MLD class, cohomology, patch frequencies), a different object from the reflection-group word-growth sequence u_d studied here, and do not address it. The shape-dependence of the minimal polynomial of ζ_T (cf. Remark 6) is the natural place a difference between shapes could appear; in §4.1 we bound this effect effectively (it cannot affect u_d unless $c_T \le 2d$); in §4.2 we show it *always* eventually does, at a depth linear in c_T.

2. The Coxeter-group identity (Fibonacci phase)

Assume T is a non-isosceles right triangle with the right angle at V_2. Then R_0 and R_1 are reflections across the two sides incident to V_2, meeting at $\pi/2$, hence $R_0R_1R_0R_1 = \mathrm{id}$. The remaining two angles are irrational multiples of π by Niven's theorem (using the non-isosceles assumption), so no relation $(R_iR_j)^m = 1$ holds for any other pair. The abstract group is

$$W = \langle\, R_0, R_1, R_2 \;\big|\; R_i^2 = 1\,(i=0,1,2),\; (R_0R_1)^2 = 1\,\rangle.$$

By Steinberg's word-growth formula the Poincaré series is $W(t) = (1+t)^2/(1-t-t^2)$, so $[t^d]W(t) = F_{d+3}$ for $d \ge 1$.

Corollary 1. *For $1 \le d \le 9$ (i.e. until the first Euclidean lattice collision at $d = 10$), $u_d = F_{d+3}$. (At $d = 0$, $u_0 = 1 \ne F_3 = 2$.)*

3. Eight length-10 affine relations

Exact-rational breadth-first search yields $u_{10} = 225$, a deficit of 8 from $F_{13} = 233$. These 8 length-10 affine relations are listed in Table 1.

#	word A	word B
1	$R_0R_1R_2R_0R_2R_0R_1R_2R_1R_2$	$R_2R_1R_2R_0R_1R_2R_0R_2R_0R_1$
2	$R_0R_1R_2R_1R_2R_0R_1R_2R_0R_2$	$R_2R_0R_2R_0R_1R_2R_1R_2R_0R_1$
3	$R_0R_2R_0R_1R_2R_1R_2R_0R_1R_2$	$R_2R_0R_1R_2R_1R_2R_0R_1R_2R_0$
4	$R_0R_2R_0R_2R_0R_1R_2R_1R_2R_0$	$R_1R_2R_1R_2R_0R_1R_2R_0R_2R_1$
5	$R_0R_2R_0R_2R_0R_1R_2R_1R_2R_1$	$R_1R_2R_1R_2R_0R_1R_2R_0R_2R_0$
6	$R_0R_2R_1R_2R_0R_1R_2R_0R_2R_0$	$R_1R_2R_0R_2R_0R_1R_2R_1R_2R_1$
7	$R_0R_2R_1R_2R_0R_1R_2R_0R_2R_1$	$R_1R_2R_0R_2R_0R_1R_2R_1R_2R_0$
8	$R_1R_2R_0R_1R_2R_0R_2R_0R_1R_2$	$R_2R_0R_1R_2R_0R_2R_0R_1R_2R_1$

Table 1: The eight length-10 affine relations. Each row's two words produce the same affine matrix in the side-reflection action of W on any right triangle.

Theorem 2 (Eight length-10 affine relations)**.** *Each of the eight identities of Table 1 holds for every right triangle. With the placement $A = (0,0)$, $B = (a,0)$, $C = (a,b)$, $a,b > 0$, the affine matrices $M_A = R_{w_A[0]} \cdots R_{w_A[9]}$ and $M_B = R_{w_B[0]} \cdots R_{w_B[9]}$ satisfy $M_A = M_B$ as 3×3 affine matrices over $\mathbb{Q}(a,b)$.*

Proof. This is a finite algebraic verification. Each reflection R_i acts as an affine isometry whose matrix entries are rational functions of a, b over $\mathbb{Q}$ (with denominators dividing a power of $a^2 + b^2$). Each entry of $M_A - M_B$ is a rational function in $\mathbb{Q}(a, b)$; clearing denominators gives a polynomial identity which is verified by SymPy in $\mathbb{Q}(a, b)$ (deterministic, no heuristics). All eight identities have additionally been *formally verified* in the Lean 4 proof assistant: the theorems `rel1_symbolic`, ..., `rel8_symbolic` in `lean/with_mathlib/SymbolicVerification.lean` each discharge the six affine-matrix-entry identities (over $\mathbb{Q}[a, b]$) defining $M_A = M_B$ by the `ring` tactic. As these proofs use only `ring`, they are checked by the Lean kernel, so the identities hold for every right triangle, not merely on a sample of specializations. □ □

Proposition 3 (Completeness at depth 10)**.** *Among the* 233 *canonical Coxeter words of length* 10 *in* W*, exactly* 225 *have pairwise distinct images under* ρ_T *for every right triangle* T*:* 217 *singleton classes and* 8 *collision pairs, each pair being one of the rows of Table 1. No further length-*10 *affine identities exist beyond the eight relations and the abstract Coxeter relations.*

Proof. Finite enumeration. All 233 canonical Coxeter words of length 10 are enumerated; the affine matrix of each is computed over $\mathbb{Q}$ on the reference $(3, 4, 5)$ triangle in exact rational arithmetic, the words are grouped by matrix, and the resulting partition has exactly 225 blocks (8 of size 2, 217 of size 1). The eight size-2 blocks are cross-checked against six additional triangles $(5, 12), (8, 15), (7, 24), (1, 2), (2, 3), (1, 7)$, and each block produces the same affine matrix in all seven cases. By Theorem 2 the eight identities hold symbolically over $\mathbb{Q}(a, b)$ and are therefore the only length-10 collisions. □ □

4. Universality across right triangles

By Theorem 2 the kernel N of $W \to \mathrm{Aff}(\mathbb{R}^2)$ contains the eight length-10 relations in addition to the Coxeter relations, and these relations hold symbolically over $\mathbb{Q}(a, b)$, so the collision structure they encode is T-independent. We record what this establishes about u_d^T, separately from the all-depths statement, which is *resolved — negatively —* in §4.2: every rational shape eventually deviates, at a depth linear in its arithmetic complexity.

Theorem 4 (Universality, established through depth 22)**.** *Let* $W = \langle R_0, R_1, R_2 \mid R_i^2 = 1,\ (R_0 R_1)^2 = 1\rangle$*, and for each right-triangle parameter* $T = (a, b)$ *with positive rational legs satisfying*

$$\boxed{a \neq b}$$

let $\rho_T : W \to \mathrm{Aff}(\mathbb{R}^2)$ *be the side-reflection realization, with* $u_d^T := |\{\rho_T(w) : |w|_W = d\}|$*. Then:*

(i) *Every* ρ_T *has the* same *linear quotient* $W \twoheadrightarrow Q$*, determined by the reflection pattern and independent of the legs; hence the translation kernel* $T_{\rho_T} = \ker(W \to Q)$ *and its abelianization* $T_{\rho_T}^{\mathrm{ab}} \cong I_A \cap I_B$ *are the same* T*-independent* $\mathbb{Z}[Q]$*-module for every such* T *(Lemma 29, Remark 32).*

(ii) *The eight length-*10 *relations of Table 1 hold over* $\mathbb{Q}(a, b)$ *for every such* T *(Theorem 2); consequently the depth-*≤ 10 *collision structure, and in particular* u_d^T *for* $0 \leq d \leq 10$*, is independent of* T*.*

(iii) u_d^T *is independent of* T *for all* $0 \leq d \leq 22$*; this is certified symbolically over* $\mathbb{Q}(a, b)$*, simultaneously for every right triangle with positive unequal legs, by the Lean 4 theorem* `universal_layers_through_22` *(Remark 8).*

Conjecture 5 (All-depths universality — now refuted)**.** *For every right-triangle parameter* $T = (a, b)$ *with positive rational unequal legs,* $\ker(\rho_T)$ *is the same subgroup of* W*, so* u_d^T *is independent of* T *for* all $d \geq 0$.

We state the conjecture in its original form for the record; it is **false**, for every T, by the explicit construction of Theorem 15 below. The correct statement is the deviation-depth sandwich of Corollary 16: u_d^T agrees with the generic sequence u_d exactly up to a first deviation depth n_T that is *linear* in the arithmetic complexity c_T of the shape.

Remark 6 (The all-depths statement)*.* Conjecture 5 is verified through depth 22 (Theorem 4(iii)) and is supported by the T-independent structural facts of (i)–(ii): the common linear quotient Q, the T-independent translation module $I_A \cap I_B$, and the symbolic depth-≤ 10 relations. These establish that $\ker(\rho_{\text{abs}}) \subseteq \ker(\rho_T)$ for every T (see the factorization below); what is *not* proven in general is the reverse inclusion $\ker(\rho_T) \subseteq \ker(\rho_{\text{abs}})$ uniformly in T, i.e. that no leg-dependent extra collision appears at some depth > 22. This reverse inclusion in fact fails for *every* T (Theorem 15); the certified content of the present section is that it holds *through depth* 30.

A concrete candidate obstruction makes the shape-dependence explicit. Chasing $\ker(\rho_T)$ through the translation module, its restriction there corresponds to the ideal of integer Laurent polynomials in $t = XY$ (the rotation generator) that vanish at the rotation number $\zeta_T = (a+bi)/(a-bi)$, a Gaussian-rational point on the unit circle. By Niven's theorem ζ_T is not a root of unity, and its minimal polynomial *depends on the shape*: e.g. $(a, b) = (1, 2)$ gives $\zeta_T = (-3 + 4i)/5$ with minimal polynomial $5t^2 + 6t + 5$, while $(a, b) = (3, 4)$ gives $\zeta_T = (-7 + 24i)/25$ with minimal polynomial $25t^2 + 14t + 25$, and the ideals these generate in $\mathbb{Z}[t^{\pm 1}]$ differ. All-depths universality therefore requires that the word-difference sets arising in the BFS never realize an element distinguishing these shape-dependent ideals; equivalently, that a structural constraint prevents such an element from appearing at any depth. This non-distinguishing is verified through depth 22 by the proof above, through depth 30 uniformly in T by the arithmetic certificate of §4.1, and through depth 30 by exact-rational BFS for the shapes $(1, 2)$ and $(3, 4)$. Theorem 10 bounds the obstruction quantitatively: a shape-dependent collision at depth d forces the shape's minimal polynomial μ_T to divide a realized word-difference, hence c_T to divide its extreme coefficients, so $c_T \leq 2d$. The obstruction *does* materialize: Theorem 15 constructs, for every T, an explicit word-difference divisible by μ_T, of length $O(c_T)$ — so the deviation is real, merely pushed beyond the reach of every computation previously performed.

Remark 7 (The isosceles case)*.* The hypothesis $a \neq b$ is essential. When $a = b$, the leg-swap symmetry adds a $\mathbb{Z}/2$ stabilizer to the orbit action, and the orbit sequence becomes $1, 3, 5, 8, 11, 13, 16, \ldots$, which is strictly smaller than the universal sequence $a(n)$ of Table 2 for $n \geq 4$. The universality phenomenon of Theorem 4 does not apply to the isosceles case. See the OEIS entry A396406 and its comments. In the framework of §4.1 the isosceles case is the extreme instance $c_T = 1$ of Theorem 15: $\zeta_T = i$ is a root of unity, the kernel of ρ_T contains the translation lattice multiple $2(t-1)(t^2+1)$, and the deviation arrives at depth 4 rather than at depth $\sim c_T$.

Remark 8 (Formal verification of universality through depth 22)*.* The T-independence of the orbit-growth sequence has been *machine-verified* for depths $0 \leq d \leq 22$ in Lean 4 (theorem `universal_layers_through_22` in `lean/with_mathlib/ComputableUniversality.lean`). That proof carries out a single breadth-first sweep over the canonical Coxeter words of W, deduplicating the symbolic affine images $\rho_{(a,b)}(w) \in \mathrm{Aff}(\mathbb{Q}(a, b))$ by their numerators relative to the common denominator $(a^2 + b^2)^{22}$, and certifies that the layer counts are exactly

$$1, 3, 5, 8, 13, 21, 34, 55, 89, 144, 225, 351, 554, 875, 1345, 2066, 3203, 4971, 7574, 11543, 17683, 27108, 41067$$

simultaneously for every right triangle with positive unequal legs. This kernel-checked certificate matches the depth reached by the independent multi-triangle reproduction of §5 and removes any reliance on a single specialization for $d \leq 22$. It establishes Theorem 4(iii); it does *not* establish the all-depths Conjecture 5.

4.1 An arithmetic effective universality theorem

The obstruction of Remark 6 can be made quantitative. The observation is that the entire side-reflection action admits a *triangle-independent symbolic normal form* in which the shape enters only through the evaluation of integer Laurent polynomials at the rotation number ζ_T; collisions then become divisibility statements in $\mathbb{Z}[i]$, and the rational root theorem yields an unconditional, all-depths universality theorem for all sufficiently "arithmetically large" triangles.

Identify $\mathbb{R}^2 = \mathbb{C}$, place the right angle at the origin with the legs along the positive axes, and rescale by $1/a$; neither operation changes word-lengths or collision structure. Writing $\sigma(z) = \bar{z}$ and $\zeta = \zeta_T = (a+bi)/(a-bi)$, the three side-reflections become

$$R_x : z \mapsto \sigma(z), \qquad R_y : z \mapsto -\sigma(z), \qquad R_h : z \mapsto \zeta^{-1}\sigma(z) + (1-\zeta^{-1}),$$

where R_x, R_y are the reflections in the leg lines and R_h in the hypotenuse line.

Lemma 9 (Symbolic normal form)**.** *For every $w \in W$ of word-length ℓ there are $\varepsilon \in \{\pm 1\}$, $\delta \in \{0,1\}$, $k \in \mathbb{Z}$ with $|k| \le \ell$, and an integer Laurent polynomial $P_w \in \mathbb{Z}[t^{\pm 1}]$,* all independent *of T, such that*

$$\rho_T(w) \colon z \;\longmapsto\; \varepsilon\,\zeta^k\,\sigma^\delta(z) \;+\; P_w(\zeta).$$

Moreover $\operatorname{supp} P_w \subseteq [-\ell,\ell]$*, every coefficient of P_w has absolute value at most ℓ, and $P_w(1) = 0$.*

Proof. Induction on ℓ. The three generators have $(\varepsilon,\delta,k,P) = (1,1,0,0)$, $(-1,1,0,0)$, and $(1,1,-1,\,1-t^{-1})$, which satisfy all the bounds and $P(1) = 0$. For the inductive step, composition gives $P_{gw} = P_g + \varepsilon_g\, t^{k_g} P_w(t^{\pm 1})$ (the sign of the exponent being -1 exactly when $\delta_g = 1$, since σ conjugates ζ to ζ^{-1} and P_w has real coefficients), and $k_{gw} = k_g \pm k_w$. Multiplying by $\pm t^{k_g}$ and inverting t change neither the coefficient bound nor the value at $t = 1$; adding the generator's P_g (height ≤ 1, $P_g(1) = 0$) increases the height by at most 1 and the support radius by at most 1, and preserves vanishing at 1. □

Since ζ_T is not a root of unity (Niven), the powers $\pm\zeta^k$ are pairwise distinct, so two group elements can have equal ρ_T-images only if their symbolic data agree in (ε,δ,k) and their translation polynomials agree at $t = \zeta_T$. Writing u_d^{sym} for the number of distinct symbolic tuples at depth d (a T-independent integer, equal to the generic count of Theorem 4), any deviation $u_e^T \ne u_e$ at some depth $e \le d$ therefore forces a *nonzero* difference $D = P_w - P_{w'} \in \mathbb{Z}[t^{\pm 1}]$ of two same-class ball elements with

$$D(\zeta_T) = 0. \tag{1}$$

Theorem 10 (Effective universality)**.** *Scale the legs to coprime integers (a,b), $a \ne b$, and write $\zeta_T = \mu/\lambda$ in lowest terms in $\mathbb{Z}[i]$ (so $\lambda = a - bi$ when $a+b$ is odd, and $\lambda = (a-bi)/(1+i)$ when a,b are both odd). Set $c_T := N(\lambda)$. Then:*

(i) c_T is odd, every prime factor of c_T is $\equiv 1 \pmod 4$, and $c_T \ge 5$; moreover $c_T = 1$ if and only if the triangle is isosceles (in which case $\zeta_T \in \{\pm i\}$ is a root of unity, and the orbit count genuinely degenerates, Remark 7).

(ii) Let $\mu_T(t) = c_T\,t^2 - e_T\,t + c_T \in \mathbb{Z}[t]$ be the primitive integer polynomial with roots $\zeta_T^{\pm 1}$ (the shape-dependent minimal polynomial of Remark 6; e.g. $5t^2 + 6t + 5$ for $(1,2)$). If $u_e^T \ne u_e$ for some $e \le d$, then μ_T divides a realized word-difference in $\mathbb{Z}[t]$; in particular c_T divides both extreme coefficients *of that difference, so $c_T \le 2d$. Consequently* every *triangle with $c_T > 2d$ satisfies universality at all depths $\le d$; equivalently, for every unequal-leg rational triangle,*

$$u_d^T = u_d \quad \textit{for all } d < \tfrac{1}{2}\,c_T\,.$$

Proof. (i) If $a+b$ is odd then $c_T = a^2 + b^2$ is odd; if a,b are both odd then $a^2 + b^2 \equiv 2 \pmod 8$ and $c_T = (a^2+b^2)/2$ is odd. A rational (inert) prime $q \equiv 3 \pmod 4$ dividing $c_T = \lambda\bar{\lambda}$ would divide λ itself, contradicting primitivity of λ (a common rational prime factor of the real and imaginary parts would divide both a and b); hence all odd prime factors split. The smallest odd integer > 1 all of whose prime factors are $\equiv 1 \pmod 4$ is 5. Finally $c_T = 1$ iff ζ_T is a unit of $\mathbb{Z}[i]$, i.e. $\zeta_T \in \{\pm 1, \pm i\}$, which for positive legs means $a = b$.

(ii) Since $a, b > 0$, ζ_T is non-real, so its minimal polynomial over $\mathbb{Q}$ has degree 2, with the conjugate root $\overline{\zeta_T} = \zeta_T^{-1}$; the primitive integer representative is $\mu_T(t) = c_T t^2 - e_T t + c_T$ with $e_T = 2(a^2 - b^2)$ when $a + b$ is odd and $e_T = a^2 - b^2$ when a, b are both odd (μ_T is primitive: an odd prime dividing c_T and e_T would divide both $a^2 + b^2$ and $a^2 - b^2$, hence both a and b). Take $D \neq 0$ as in (1), built from two ball elements of depth $\leq d$; by Lemma 9 its coefficients are bounded by $2d$ in absolute value. Normalize $\widetilde{D} = t^m D \in \mathbb{Z}[t]$ with $\widetilde{D}(0) \neq 0$. Since $\widetilde{D}$ has integer coefficients and vanishes at ζ_T, Gauss's lemma gives $\widetilde{D} = \mu_T \cdot E$ with $E \in \mathbb{Z}[t]$. Comparing top and bottom coefficients, $\mathrm{lead}(\widetilde{D}) = c_T \,\mathrm{lead}(E)$ and $\mathrm{const}(\widetilde{D}) = c_T \,\mathrm{const}(E)$, so c_T divides both extreme coefficients of $\widetilde{D}$ and $c_T \leq |\mathrm{lead}(\widetilde{D})| \leq 2d$. □

Theorem 10(ii) is, to our knowledge, the first *all-depths* universality statement: for each fixed depth it reduces the conjecture to the finitely many shapes with $c_T \leq 2d$ — by (i) these are very sparse: at depth 30 the splitting condition leaves only $c_T \in \{5, 13, 17, 25, 29, 37, 41, 53\}$ (16 rotation numbers in all; e.g. $45 = 3^2 \cdot 5$ and $49 = 7^2$ are excluded because 3 and 7 are inert) — and those can be checked mechanically. Evaluating the entire symbolic ball at each candidate ζ in the finite field $\mathbb{F}_q$ (choosing $q \equiv 1 \pmod 4$ so that $i \in \mathbb{F}_q$, and $q \nmid c_T$) is *one-sided exact*: images distinct mod q are distinct in $\mathbb{Q}(i)$. Running this certificate at depth 30 (ball of 3,336,511 symbolic elements, 16 candidate rotation numbers with $c_T \leq 60$, two independent 31-bit primes; script `code/zeta_probe/certify.py`, total runtime under three minutes) found every candidate collision-free. Combined with Theorem 10(ii) for $c_T > 60$ this proves:

Theorem 11 (Sharp uniform universality: depth 32)**.** *For every right triangle with positive unequal rational legs, $u_d^T = u_d$ for all $0 \leq d \leq 32$. The bound is* sharp*: the $(1, 2)$ triangle satisfies $u_{33}^{(1,2)} = u_{33} - 30 = 3{,}848{,}354 < u_{33}$, so depth 32 is the exact threshold below which the orbit count is shape-independent.*

The certificate was run in Python/NumPy at depth 30 (`certify.py`) and re-implemented in Rust (`certify38_rust/`) through depth 36: the generic layer counts match the published values through depth 36, all 22 candidate rotation numbers with $c_T \leq 72$ are collision-free through depth 32, and the single collision beyond is the $(1, 2)$ deviation quantified below. The trust base is exact modular arithmetic with two independent 31-bit primes rather than the Lean kernel; it extends the kernel-checked Lean bound of Remark 8 by ten further layers.

4.2 Resolution of the conjecture: every algebraic shape eventually deviates

The obstruction of Remark 6 does materialize — for *every* algebraic shape, at an explicitly bounded depth. The mechanism is elementary Euclidean geometry: because the two legs meet at a right angle, $R_x R_y$ is the half-turn about the right-angle vertex, so $R_x R_y R_h$ is a *glide reflection* along the hypotenuse — and the square of a glide reflection is a pure translation.

Lemma 12 (Glide-square translation lattice)**.** *In the symbolic group, $\tau := (R_x R_y R_h)^2$ is the pure translation by $2(t^{-1} - 1)$, i.e. the tuple $(1, 0, 0, 2(t^{-1} - 1))$. Writing $r := R_x R_h$ (rotation, linear part t) and $\mathcal{T}$ for the subgroup of pure translations of the symbolic group, conjugates $r^k \tau^m r^{-k}$ are the translations by $2m\, t^k (t^{-1} - 1)$, and products of these realize*

$$2(t - 1)\, \mathbb{Z}[t^{\pm 1}] \;\subseteq\; \mathcal{T}.$$

Proof. $R_x R_y \colon z \mapsto -z$, so $g := R_x R_y R_h \colon z \mapsto -\zeta^{-1}\bar{z} - (1 - \zeta^{-1})$, and $g^2(z) = -\zeta^{-1}\,\overline{g(z)} - (1 - \zeta^{-1}) = z + 2(\zeta^{-1} - 1)$, an identity of symbolic tuples (it uses only $\zeta\bar{\zeta} = 1$). Conjugating a translation by the rotation r^k multiplies its vector by t^k; translations commute, so products realize $2(t^{-1} - 1) f(t)$ for every $f \in \mathbb{Z}[t^{\pm 1}]$. □

In fact the glide-square lattice is *everything*: the inclusion of Lemma 12 is an equality.

Theorem 13 (The translation lattice)**.** *The pure-translation subgroup of the symbolic group is exactly*

$$\mathcal{T} \;=\; 2(t-1)\,\mathbb{Z}[t^{\pm 1}].$$

Proof. After Lemma 12 only $\mathcal{T} \subseteq 2(t-1)\mathbb{Z}[t^{\pm1}]$ remains. Reduce the symbolic group modulo 2: the reduction $(\varepsilon, \delta, k, P) \mapsto ((k,\delta),\, P \bmod 2)$ is a homomorphism onto a subgroup of $\mathbb{F}_2[t^{\pm1}] \rtimes (\mathbb{Z} \rtimes \mathbb{Z}/2)$, because the sign ε acts trivially on coefficients mod 2. Under it the two leg reflections — which differ only by the sign — acquire the *same* image $s = ((0,1),0)$, while the hypotenuse reflection maps to $\bar{h} = ((-1,1),\, 1 + t^{-1})$. Both are involutions ($\bar{h}^2$ has translation $(1+t^{-1}) + t^{-1}(1+t) = 0$ over $\mathbb{F}_2$), so the image of the whole symbolic group is generated by *two involutions* and is therefore dihedral: every element is $(s\bar{h})^n$ or $(s\bar{h})^n s$ up to inversion. Since $s\bar{h}$ has linear part $(1,0)$ — the rotation t, of infinite order — the only image element with trivial linear part is the identity, whose translation is 0. A pure translation of the symbolic group has trivial linear part, so its image is the identity, i.e. all its coefficients are even: $\mathcal{T} \subseteq 2\,\mathbb{Z}[t^{\pm1}]$. Combined with $P(1) = 0$ (Lemma 9), every $P \in \mathcal{T}$ is an even multiple of $(t-1)$. □

Mod 2, the two perpendicular legs collapse into a single mirror and the generic group degenerates to the infinite dihedral group, which has no translations; this is what caps the lattice. Empirically, all 130 pure translations in the symbolic 16-ball have every coefficient even, as the theorem demands.

Corollary 14 (Complete kernel classification)**.** *Let T have positive unequal legs with algebraic ratio, and let μ_T be the primitive integer minimal polynomial of ζ_T (self-reciprocal, of degree 2 in the rational-leg case). Then*

$$\ker \rho_T \;=\; \{\textit{translations by } 2(t-1)\,\mu_T\, g : g \in \mathbb{Z}[t^{\pm1}]\} \;\cong\; \mathbb{Z}[t^{\pm1}],$$

and $\ker \rho_T$ *is the normal closure of the* single *witness element w_T of Theorem 15: every algebraic specialization of the generic group is a one-relator (normally generated) quotient. Moreover any collision difference is divisible by $2(t-1)\mu_T$, whose extreme coefficients are $\pm 2c_T$; by the height bound this* doubles *the lower bound of Corollary 16:*

$$n_T \;\geq\; c_T.$$

Proof. By Niven a kernel element has trivial linear part, hence is a pure translation $P \in \mathcal{T}$ with $P(\zeta_T) = 0$; writing $P = 2(t-1)g$ (Theorem 13) and using $\zeta_T \neq 1$, this is equivalent to $\mu_T \mid g$. Conjugation acts on translations by $\pm t^k$ and $P \mapsto P(t^{-1})$; since μ_T is self-reciprocal ($t^2\mu_T(t^{-1}) = \mu_T(t)$), the $\mathbb{Z}[t^{\pm1}]$-module generated by the single translation $2(t-1)\mu_T$ is stable and exhausts the kernel. For the bound: a deviation at depth d produces a collision difference D of height $\leq 2d$ divisible by $2(t-1)\mu_T$, whose leading coefficient is $2c_T \cdot \mathrm{lead}(g) \neq 0$, so $2c_T \leq 2d$. □

The lattice $2(t-1)\mathbb{Z}[t^{\pm1}]$ is an *ideal*: it contains a multiple of every integer polynomial — in particular of every shape's minimal polynomial μ_T. That single observation refutes the conjecture.

Theorem 15 (Non-universality: resolution of Conjecture 5)**.** *Conjecture 5 is false — for every unequal-leg rational triangle. Explicitly, with $\mu_T(t) = c_T t^2 - e_T t + c_T$ as in Theorem 10, the word*

$$w_T \;=\; \tau^{\,c_T}\, r\, \tau^{\,-e_T}\, r\, \tau^{\,c_T}\, r^{-2}, \qquad \tau = (R_x R_y R_h)^2, \quad r = R_x R_h,$$

is a nontrivial element of the symbolic group — the pure translation by $2(t^{-1}-1)\,\mu_T(t) \neq 0$ — of word-length $6(2c_T + |e_T|) + 8 \leq 24\,c_T + 8$, whereas $\rho_T(w_T)$ is the identity *isometry: its translation vector is $2(\zeta_T^{-1} - 1)\,\mu_T(\zeta_T) = 0$. Hence* $\ker(\rho_T) \supsetneq \ker(\rho_{\mathrm{abs}})$*, and $u_d^T < u_d$ for some $d \leq 6(2c_T + |e_T|) + 8$.*

More generally, for legs $(1, \beta)$ with β any positive algebraic number $\neq 1$, the same construction applied to the primitive integer minimal polynomial of ζ_T produces a nontrivial kernel element.

Proof. By Lemma 12 the word w_T is the symbolic translation by $2(t^{-1}-1)\,(c_T - e_T\,t + c_T\,t^2) = 2(t^{-1}-1)\,\mu_T(t)$, a nonzero Laurent polynomial, so $w_T \neq 1$ in the symbolic group; its letter count is $6c_T + 2 + 6|e_T| + 2 + 6c_T + 4$. Under ρ_T the translation vector is the evaluation at $t = \zeta_T$, which vanishes because $\mu_T(\zeta_T) = 0$. A nontrivial kernel element of length ℓ forces $|\rho_T(B_\ell)| < |B_\ell|$ for the symbolic ℓ-ball, hence a strictly smaller layer count at some depth $\leq \ell$. The witness has been verified in exact rational arithmetic for $(a, b) = (1, 2)$ (length 104), $(3, 4)$ (length 392), and $(1, 3)$ (length 116): `code/zeta_probe/witness.py`. □

Corollary 16 (Deviation-depth sandwich)**.** *For every unequal-leg rational triangle let $n_T := \min\{d : u^T_d \neq u_d\}$, which is finite by Theorem 15. Then*

$$\max(33,\ c_T) \ \leq\ n_T\ \leq\ 6\,(2c_T + |e_T|) + 8\ \leq\ 24\,c_T + 8,$$

the lower bounds by Corollary 14 and Theorem 11, the upper bound by the witness word. The first deviation depth is pinned to linear growth in the arithmetic complexity c_T of the shape. The bounds are realistic: for $(1, 2)$ they read $33 \leq n_{(1,2)} \leq 104$, and the measured value is $n_{(1,2)} = 33$ — the certified lower bound is attained. (By the depth-38 certificate the lower bound improves to $\max(39, c_T)$ *for every shape other than $(1, 2)$.)*

Theorem 17 (The deviation depth is a shortest-vector length)**.** *For every unequal-leg rational triangle,*

$$n_T\ =\ \left\lceil \tfrac{1}{2}\,L_T \right\rceil, \qquad L_T\ :=\ \min_{0 \neq f \in \mathbb{Z}[t^{\pm 1}]} \ell\big(\textit{translation by } 2(t-1)\,\mu_T\,f\big),$$

the length of the shortest nontrivial element of $\ker \rho_T$ *in the word metric of Theorem 23.*

Proof. By Corollary 14 the nontrivial kernel elements are exactly the translations $2(t-1)\mu_T f$, $f \neq 0$. If $w \in \ker \rho_T$ has length L, write $w = g_1 g_2$ along a geodesic with $\ell(g_1) = \lceil L/2 \rceil$ and $\ell(g_2) = \lfloor L/2 \rfloor$. Then g_1 and g_2^{-1} are distinct elements of the ball of radius $\lceil L/2 \rceil$ (the generating set is symmetric) with $g_1 = w\,g_2^{-1}$, so they map to the same isometry — a collision inside the ball of radius $\lceil L/2 \rceil$, so $n_T \leq \lceil L_T/2 \rceil$. Conversely a deviation at depth d produces two distinct ball-d elements with equal image, whose quotient is a kernel element of length $\leq 2d$; hence $L_T \leq 2n_T$, i.e. $n_T \geq \lceil L_T/2 \rceil$. The minimum is over a finite effective set: the metric formula bounds ℓ from below by the coefficient sum of $2(t-1)\mu_T f$, which grows with the degree and height of f. □

Remark 18 (The invariant n_T, computed and predicted). Theorem 17 converts the first-deviation depth into a *shortest-vector problem in the word metric* — the analogue, for this lamplighter metric, of a shortest-lattice-vector computation, and it exhibits the classical shortest-vector phenomenon: *the shortest kernel element is not the obvious generator.* Extracting the kernel polynomials from all 30 exactly-verified depth-33 collision pairs of the $(2, 1)$ shape shows that every one uses the multiplier $f = \pm t^j(1+t)$ — never $f = 1$. The reason is coefficient cancellation: $(t^2 - 1)\mu_T$ has coefficient sum 22 against 32 for $(t-1)\mu_T$, and indeed a meet-in-the-middle computation over the radius-34 ball ($L = \min_g d(g) + d(g^{-1}w)$ over 17,214,686 ball elements) certifies $L_{(2,1)} = 66$ exactly, attained by $f = t^{-4}(1 + t)$, while the $f = 1$ element has length > 68. Thus $n_{(2,1)} = \lceil 66/2 \rceil = 33$ is *re-derived from first principles*, in exact agreement with the measured deviation depth — the invariant n_T is now computed, not merely observed. The optimal multiplier obeys the shape mirror: $(1, 2)$, with the reflected minimal polynomial $5t^2 + 6t + 5$, has $L = 66$ attained by $f = 1 - t$ (the $t \mapsto -t$ image of $(2, 1)$'s $f = 1 + t$), so $n_{(1,2)} = 33$ as well. We record as the computable, falsifiable refinement of the sandwich: the certificate data already separates the two sign patterns, since the shape $(1, 3)$ ($e_T = +8$, alternating deposits) satisfies $n_{(1,3)} > 38$, strictly later than the all-positive pattern of $(1, 2)$ would suggest.

Corollary 19 (Transcendence dichotomy). *For a right triangle with legs $(1, \beta)$, $\beta > 0$, $\beta \neq 1$: the representation ρ_T is faithful on the symbolic group — equivalently, $u_d^T = u_d$ for* all $d \geq 0$ *— if and only if β is transcendental.*

Proof. $\zeta_T = (1 + i\beta)/(1 - i\beta)$ is algebraic iff β is. If β is transcendental, a kernel element would by Lemma 9 and the argument before (1) be a pure translation with a nonzero integer Laurent polynomial vanishing at ζ_T — impossible. If β is algebraic, multiply the primitive integer minimal polynomial of ζ_T by the lattice of Lemma 12 as in Theorem 15. ☐

Thus the "universal" sequence u_d is precisely the orbit-growth sequence of the *generic* right triangle: it is realized exactly by the (uncountably many) shapes with transcendental leg ratio, while every rational — indeed every algebraic — shape agrees with it through depth $\approx c_T$ and then falls strictly below, the isosceles square-triangle ($c_T = 1$, deviation at depth 4) being the extreme case of the same phenomenon. Every computation previously reported is consistent with the sandwich: all certified and BFS-verified ranges lie below the deviation depths of the shapes used.

Remark 20 (A height-graded family; outlook). The per-shape statements of this section package into a single map $\zeta \mapsto W_{\text{gen}}/\langle\langle w_\zeta \rangle\rangle$ from algebraic numbers on the unit circle to one-relator quotients of the generic group: by Corollary 14 the kernel for rotation number ζ is the principal ideal $2(t-1)\mu_\zeta(t)\,\mathbb{Z}[t^{\pm 1}]$ of the lamplighter translation lattice, where μ_ζ is the (self-reciprocal) minimal polynomial of ζ, and the deviation depth $n_\zeta = \lceil L_\zeta/2 \rceil$ is the lamplighter shortest-vector length of that ideal (Theorem 17). This holds verbatim for ζ of any degree, so the whole picture extends from rational right triangles (μ_ζ quadratic) to arbitrary algebraic leg ratios, graded by the arithmetic complexity of ζ. The shortest vector L_ζ is controlled by a *height* of $2(t-1)\mu_\zeta$ between its ℓ^1-norm h_1 and its (logarithmic) Mahler measure Mah. Computing L_ζ (in the relaxed metric, a proven upper bound, equal to the true length on the certified shapes $c_T = 5$ and $(1, 3)$) across rational shapes and the first two algebraic ratios — $\beta = \sqrt{2}$, with $\mu = 3t^2 + 2t + 3$, and $\beta = \varphi$, with $\mu = 5t^4 + 6t^2 + 5$ — gives the sandwich

$$\text{Mah} \;\leq\; h_1 \;\leq\; L_\zeta \;\leq\; 3\,h_1 \qquad \text{(all eight tabulated shapes)},$$

and, for every *quadratic* μ (including $\sqrt{2}$), the clean exact law

$$L_\zeta \;=\; \tfrac{3}{2}\,h_1 + 3\,\text{Mah} \qquad \Big(\text{e.g. } (1,2)\colon \tfrac{3}{2}\cdot 24 + 3\cdot 10 = 66;\quad \sqrt{2}\colon \tfrac{3}{2}\cdot 16 + 3\cdot 6 = 42\Big).$$

The law *breaks* at φ — where the shortest vector needs the cancelling multiplier $f = 1 + t$, not a monomial, giving $L_\varphi = 78$ against the formula's 126. Thus the closed form holds exactly while μ is quadratic and fails precisely when $\deg \mu \geq 4$ admits coefficient cancellation; the shortest vector is genuinely not the naive generator, and n_T is *not* linear in (c_T, e_T) (failing e.g. for $(2, 3)$ and $(3, 4)$). We flag the height law and its arithmetic break as the most natural generalization of the present results (computed exactly in `code/zeta_probe/route_c_height_family.py`; the sandwich and the law are conjectural for the connectivity-true metric beyond the two certified shapes).

Data provenance. The depth-38 certificate (`certify38_rust/`, independently reproduced on a second machine; 87,838,773-element symbolic ball, all 24 candidate rotation numbers with $c_T \leq 76$) settles the provenance question in full: the published terms $u_{31}, \ldots, u_{38}$ of A396406, originally computed by exact BFS of the $(3, 4)$ triangle, *are* the generic terms — $n_T > 38$ for every unequal-leg rational triangle except $(1, 2)$, whose deviation at $n_{(1,2)} = 33$ is the unique one in this range and is exactly quantified above.

The predicted deviation has been observed. Theorem 15 predicted $n_{(1,2)} \in [31, 104]$; the certificate finds

$$n_{(1,2)} = 33:$$

the $(1,2)$ triangle agrees with the generic sequence through depth 32 and merges exactly 30 pairs of generic elements at depth 33 ($u_{33}^{(1,2)} = 3{,}848{,}354$ versus $u_{33} = 3{,}848{,}384$), the deficit growing to 123 by depth 34 and 1,159 by depth 36 (each verified with two independent 31-bit primes). The other $c_T = 5$ shape $(1,3)$ — with the *different* minimal polynomial $5t^2 - 8t + 5$ — remains collision-free through depth 36, confirming that the deviation depth is governed by the shape's minimal polynomial, not merely by c_T. The conjecture was thus falsifiable by direct BFS all along: the first deviation sits at depth 33, just beyond the depth-30 range to which the $(1,2)$ shape had ever been pushed.

4.3 The generic group is virtually the wreath product $\mathbb{Z} \wr \mathbb{Z}$

Theorem 13 identifies the abstract isomorphism type of the generic group, and it turns out to be a central object of geometric group theory.

Theorem 21 (Lamplighter structure)**.** *Let W_{gen} denote the symbolic (generic) side-reflection group. The set H of elements whose linear part is a pure rotation t^k is a normal subgroup of index 4, and*

$$H \;=\; \mathcal{T} \rtimes \langle r\rangle \;\cong\; \mathbb{Z}[t^{\pm 1}] \rtimes_t \mathbb{Z} \;=\; \mathbb{Z} \wr \mathbb{Z}, \qquad r = R_x R_h,$$

the wreath product of $\mathbb{Z}$ by $\mathbb{Z}$. Hence W_{gen} — equivalently, the side-reflection group of every *right triangle with transcendental leg ratio (Corollary 19) — is virtually $\mathbb{Z} \wr \mathbb{Z}$. In particular it is:*

- *(i) metabelian-by-finite, hence* amenable*, while of exponential growth — the coexistence for which $\mathbb{Z} \wr \mathbb{Z}$ is the textbook example, here realized by three mirrors of a Euclidean triangle;*
- *(ii)* not finitely presentable*: by Baumslag's theorem (G. Baumslag,* Wreath products and finitely presented groups*, Math. Z. 75 (1961), 22–28) the wreath product $A \wr B$ with $A \neq 1$ is finitely presented only when B is finite, and finite presentability is a commensurability invariant.*

Proof. H is the kernel of the composite $W_{\text{gen}} \twoheadrightarrow \widetilde{Q} \to \widetilde{Q}/\langle t\rangle$, where $\widetilde{Q} = \{\pm t^k \sigma^\delta\}$ is the linear-part quotient; $\langle t\rangle$ is normal in $\widetilde{Q}$ (the sign is central and $\sigma t \sigma^{-1} = t^{-1}$) of index 4, so H is normal of index 4. Any element of H with linear part t^k differs from r^k by a pure translation, so $H = \mathcal{T}\langle r\rangle$ with $\langle r\rangle \cap \mathcal{T} = 1$, an internal semidirect product. By Theorem 13, $f \mapsto 2(t-1)f$ is a $\mathbb{Z}[t^{\pm 1}]$-module isomorphism $\mathbb{Z}[t^{\pm 1}] \to \mathcal{T}$, and conjugation by r acts on $\mathcal{T}$ as multiplication by t; since $\mathbb{Z}[t^{\pm 1}] = \bigoplus_{k \in \mathbb{Z}} \mathbb{Z}\, t^k$ with t acting as the shift, the semidirect product is precisely $\mathbb{Z} \wr \mathbb{Z}$. Amenability follows since $\mathcal{T}$ is abelian and $\widetilde{Q}$ is virtually abelian, so W_{gen} is metabelian-by-finite; exponential growth is classical for $\mathbb{Z} \wr \mathbb{Z}$ and passes to commensurable groups (consistently, by Milnor–Wolf a solvable group of non-polynomial growth must have exponential growth). □

Three consequences reframe the whole subject.

(a) The algebraic shapes are the metabelian approximations. For rational T the specialized group $\rho_T(W)$ is virtually $\mathbb{Z}[\zeta_T^{\pm 1}] \rtimes_{\zeta_T} \mathbb{Z}$, a finitely generated metabelian affine group (a relative of the solvable Baumslag–Solitar groups); the isosceles case $c_T = 1$ degenerates to the crystallographic wallpaper group. The one-parameter family of right triangles thus interpolates

$$\text{wallpaper } (c_T = 1) \;\longrightarrow\; \text{solvable affine } (c_T < \infty) \;\longrightarrow\; \text{lamplighter (transcendental ratio)},$$

and every right-triangle side-reflection group, of any shape, is amenable.

(b) The deviation theory is a quantified convergence of marked groups. The sandwich of Corollary 16 says exactly that the Cayley ball of $\rho_T(W)$ (reflection generators) is isometric to that of W_{gen} up to radius $n_T \asymp c_T$: the rational-triangle groups converge to the virtually-lamplighter group in the space of marked groups, with *exactly linear* rate in the arithmetic complexity

of the shape, and with the first approximation girth measured ($n_{(1,2)} = 33$). This places the present family inside the program of quantifying residual finiteness and group approximations (cf. K. Bou-Rabee, *Quantifying the residual finiteness of groups*, J. Algebra 323 (2010), 729–737) with completely explicit constants.

Theorem 22 (Normal form and lamp model)**.** *An abstract tuple $(\varepsilon, \delta, k, P)$ lies in W_{gen} if and only if*

$$P(1) = 0 \qquad \textit{and} \qquad P \;\equiv\; (1+t)\big(1+t+\cdots+t^{|k|-1}\big)\, t^{\min(k,0)} \pmod 2.$$

(The mod-2 class is the dihedral cocycle of Theorem 13; it depends only on k.) In particular the word and membership problems are solved explicitly. Moreover, writing $P = \sum_j a_j\,(t^{j+1} - t^j)$ in the edge basis, right multiplication by the generators acts as an edge lamplighter *on $\mathbb{Z}$:*

$$R_x\colon \delta \mapsto 1-\delta, \qquad R_y\colon (\varepsilon,\delta) \mapsto (-\varepsilon, 1-\delta),$$

and R_h moves the walker one site (down if $\delta = 0$, up if $\delta = 1$), deposits $+\varepsilon$ resp. $-\varepsilon$ on the traversed edge, and flips δ. Consecutive moves therefore bounce by default, straight travel and deposit-sign changes being paid for in R_x, R_y letters.

Proof. Necessity of the congruences is Lemma 9 and Theorem 13 (the mod-2 image of an element is determined by its image in the dihedral quotient, hence by k; the representative r^k has $P_{r^k} = (1-t)(1+\cdots+t^{k-1})$). Sufficiency: the realized translation polynomials with a fixed linear part form a single coset of $\mathcal{T}$, and the two congruences cut out exactly one such coset, since a difference satisfying both lies in $2\mathbb{Z}[t^{\pm1}] \cap (t-1)\mathbb{Z}[t^{\pm1}] = 2(t-1)\mathbb{Z}[t^{\pm1}] = \mathcal{T}$. The lamp dynamics is the composition law read in the edge basis; as a check, a breadth-first search of the lamp model reproduces $u_0, \dots, u_{14}$ exactly and every ball element satisfies the congruences. □

Theorem 23 (Metric formula)**.** *The word length of $g = (\varepsilon^*, \delta^*, k^*, a)$ with respect to the three reflections is the value of the following explicit finite optimization. Choose crossing numbers $m_j \ge |a_j|$ with $m_j \equiv a_j \pmod 2$, chained so that the support is reachable from the start site; the walk directions force $u_j = (m_j + f_j)/2$ up-crossings ($f_j = \pm 1$ on the travel interval between 0 and k^*, else 0). At minimal crossing numbers each crossing's sign is forced ($\varepsilon = -d \cdot \operatorname{sign}(a_j)$ for direction d). Each site pairs arrivals with departures: a pass (same direction) costs 1, a bounce 0, and a sign change is free at a pass but costs 2 at a bounce, with virtual boundary moves (arrival at the origin: up, $+$; departure at k^*: direction δ^*, sign ε^*); the pairing must be Eulerian-connected — an isolated cycle is inadmissible, and splicing one in costs exactly 2. Then*

$$\ell(g) \;=\; \min\Big(\textstyle\sum_j m_j + \sum_{sites} \textit{pairing costs}\Big).$$

This formula has been verified to agree with the true word metric on every *element of the radius-14 ball (3,723 elements, zero mismatches;* `code/zeta_probe/lamp_profile.py`*); moreover, enumerating the elements from the normal form of Theorem 22 and counting them by the lengths this formula assigns reproduces $u_0, \dots, u_{22}$ of A396406 exactly, with no breadth-first search anywhere in the pipeline (*`code/zeta_probe/fire_rust/`*, independently run on a second machine); the relaxation without the connectivity constraint is exact through radius 9 and undershoots by exactly 2 per isolated cycle beyond (gap histogram $\{+2\colon 70,\ +4\colon 2\}$ at radius 14).*

Proposition 24 (Growth-series complexity, empirical)**.** *Over the 43 exact terms $u_0, \dots, u_{42}$ (the first 39 published, $u_{39}, \dots, u_{42}$ from the depth-42 BFS, all re-derived from the metric formula of Theorem 23 through u_{22}):*

- *(i) the sequence satisfies* no *constant-coefficient linear recurrence of order ≤ 21 (the maximal order testable on 43 terms with positive margin); hence the growth series, if rational, has denominator degree > 21;*

(ii) *it satisfies* no *holonomic (polynomial-coefficient) recurrence within the box* (order+1)(deg +1) $\le$ 32 *(margins* 4 *to* 40*); hence the series, if D-finite, is not of low complexity;*

(iii) *it satisfies* no *algebraic equation* $P(x, F) = 0$ *with* $\deg_F P \le 6$ *and* $\deg_x P \le 14$ *(every one of* 29 *tested* $(\deg_x, \deg_F)$ *cells gives a full-column-rank coefficient matrix, over-determined by* 7 *to* 34 *equations); hence* F*, if algebraic, has high degree.*

All three are exact fraction-free nullspace searches; the same machinery recovers the Fibonacci recurrence and generating-function relation and the Catalan holonomic recurrence on positive controls, so the negatives are genuine and not detector artifacts. Naive extrapolation of the ratios u_{d+1}/u_d *is unreliable here — the convergence is* $O(1/d)$ *with a period-*4 *sign oscillation, so the* 43*-term estimate* 1.4995 *is a finite-depth* overshoot *— but the closed-form singularity analysis below (Proposition 26; Theorem 27) pins the limit:* $\beta_2 \le 1.49161778711\ldots$*, the exact growth rate of the relaxed model, which* excludes *the value* $3/2$*.*

The determinization of the metric of Theorem 23 hinges on whether the no-isolated-cycle (Eulerian-connectivity) constraint admits a bounded local encoding. It does — but *not* as a bound on the number of open interface components, which is unbounded on geodesics: the family $g_n = (\varepsilon^*, \delta^*, k^*) = (1, 0, 0)$ with single deposit $a_0 = 2n$ has geodesic word $(R_y R_h R_x)^{2n}$ of length $6n$ and forces $n + 1$ simultaneously open components crossing edge 0. The correct bound is on component *shapes*.

Lemma 25 (Strand bound)**.** *In any realization of an element as a single Eulerian trail on the crossing multigraph (vertices the integer sites;* m_j *parallel edges across the gap at site* j*), every interface component at every edge* j *carries at most one up-crossing and at most one down-crossing of edge* j*. Hence each component is one of exactly eight shapes, and an interface profile is a multiset over this fixed eight-letter alphabet whose only unbounded datum is the multiplicity of the through-shape.*

Proof. An interface component at edge j is the footprint, on the cut between sites $\le j$ and sites $> j$, of a maximal sub-walk of the trail lying in the half-line $\{\text{ site} \le j\}$. Such a sub-walk enters the half-line by crossing the gap at j downward and leaves by crossing it upward; a *second* upward crossing is impossible, since after crossing up the walk sits at site $j + 1 > j$ and has already left the half-line, ending the sub-walk. So each component has exactly one up- and one down-crossing of edge j (a marker sub-walk, beginning or ending inside, contributes one). The bound is therefore an invariant of single-trail admissibility, *independent* of length-minimization. The eight shapes are the contents $(1, 0), (0, 1), (1, 1)$ refined by the two lamp signs and the start/end marker; a strand-bounded transfer DP enforcing the bound reproduces the exact word metric with zero mismatches through radius 12. □

By Lemma 25 the edge-by-edge transfer determinizes over multisets of eight bounded shapes, the single unbounded coordinate being the crossing count of the current edge — a genuine *one catalytic variable.* Assembling the transfer explicitly in the relaxed model (which drops connectivity and over-counts by exactly +2 per isolated cycle), the per-site cost collapses to $\max(|a_{j-1}|, |a_j|)$ and the bulk generating function $F(q, t)$ ($q = x^2$, t marking the half-deposit of the current edge) satisfies the explicit, series-validated functional equation

$$F(q,t) = \frac{2qt}{1-qt} + \frac{2qt}{1-qt}\left[F(q,q) - F(q,q^2t)\right] + \frac{2q^2t}{1-q^2t}\,F(q,q^2t)$$

(verified to order x^{28}; full derivation, the local-cost lemma Site $= \max(|a_{j-1}|, |a_j|)$, and the closed-form relaxed length in `code/zeta_probe/route_b/`). This is the load-bearing structural finding, and it is *negative*: the catalytic variable enters through the *dilation* $t \mapsto q^2t$, so the "section" $F(q, q^2t)$ still depends on t. The equation is therefore a linear *q-difference* (dilation) equation, *not* a polynomial-kernel equation; the theorem of Bousquet-Mélou and Jehanne (*Polynomial equations with one catalytic variable, algebraic series and map enumeration*, J. Combin. Theory Ser. B 96 (2006), 623–672) does not apply, there is no finite kernel to cancel, and the solution is q-holonomic — generically *transcendental* over $\mathbb{Q}(x)$. The dilation is forced by the max-coupling, which is intrinsic to the metric.

The same structure also *pins the growth rate*. Unfolding the q-difference equation along $G_k := F(q, q^k)$ gives the two-step recurrence $G_k = \alpha_k(1+G_1)+\gamma_k G_{k+2}$ with $G_\infty = 0$, which telescopes to the closed form $F(q,t)\big|_{\text{bulk}} = S_0(q)/(1 - S_1(q))$ for explicit, rapidly convergent q-series S_k. The full relaxed generating function is the analogous assembly of this bulk block with a *travel* block (the $f = \pm 1$ edges, same max-coupling) whose telescoping $G_0^T = \Sigma_0/(1 - \Sigma_1)$ carries the displacement sum.

Proposition 26 (The asymptotic ratio)**.** *The relaxed growth rate is* $\beta_2^{\text{rel}} = 1/\sqrt{q^*}$*, where* $q^* = 0.4494536305589\ldots$ *is the unique small positive root of the explicit travel-block equation* $\Sigma_1(q) = 1$ *(the travel pole dominates the bulk pole* 0.60957*, and the bulk dressing,* $\Sigma_1^{\text{bulk}}(q^*) = 0.4804 < 1$*, leaves it fixed):*

$$\beta_2^{\text{rel}} = 1.4916177871143742268\ldots$$

Moreover the true *asymptotic ratio equals this value* exactly*:*

$$\beta_2 = \beta_2^{\text{rel}} = 1.4916177871143742268\ldots$$

This is not merely an upper bound. The travel block is connectivity-invariant*: a pure-translation run on sites* $0, \ldots, k$ *has odd slot-multiplicities* $m_j = |a_j| \geq 1$*, so the associated path-multigraph is connected with odd-degree vertex set exactly* $\{0, k\}$*, whence by Euler's theorem it carries a single Eulerian trail and incurs no connectivity penalty (true length = relaxed length on every travel run; verified on* 1.9×10^7 *runs,* 0 *failures). Consequently the true series* U *shares the relaxed travel block* $\Sigma_0/(1 - \Sigma_1)$ verbatim*, and its pure-travel subcount* $t_d \leq u_d$ *already grows at* β_2^{rel} *(the dominant pole* q^* *is genuine,* $\Sigma_0(q^*) = 1.156 \neq 0$*). With* $t_d \leq u_d \leq v_d$ *both flanks of the same rate* β_2^{rel}*, the squeeze gives* $\beta_2 = \beta_2^{\text{rel}}$ *exactly — pinning the long-deferred true growth rate, with no dependence on the cosine lemma* (2)*. In particular*

$$\beta_2 = 1.4916177871\ldots < \tfrac{3}{2},$$

so $3/2$ *— which the* 43*-term ratio estimate* 1.4995 *had suggested — is* excluded*; the apparent proximity was a finite-depth overshoot of an* $O(1/d)$*, sign-oscillating convergence. An integer-relation search at* 130 *digits finds no algebraic relation for* β_2^{rel} *or* q^* *of degree* ≤ 6*, consistent with the q-difference (transcendental) structure; transcendence of the* number β_2 *remains open.*

The same closed form yields *transcendence*, modulo one explicit analytic lemma. Recall $\alpha_k = 2q^{k+1}/(1 - q^{k+1})$ and $\gamma_k = \alpha_{k+1} - \alpha_k$, so $S_1 = \sum_{j \geq 0} \alpha_{1+2j} \prod_{i<j} \gamma_{1+2i}$, and likewise Σ_1 for the travel block.

Theorem 27 (Transcendence of the relaxed series, modulo Lemma 2)**.** *The relaxed generating function* $V(x) = \sum_d v_d x^d$ *of A396406, and each of its blocks* $S_0/(1 - S_1)$ *and* $\Sigma_0/(1 - \Sigma_1)$*, is transcendental over* $\mathbb{Q}(x)$*.*

Proof sketch (full proof in `code/zeta_probe/route_b/transcendence.tex`*).* Each S_k is holomorphic on $|q| < 1$: its jth term is $O(r^{2j^2})$ on $|q| \leq r < 1$ (super-geometric), so $S_0/(1 - S_1)$ is meromorphic there. On $(0, 1)$, $\alpha_k > 0$ but $\gamma_k < 0$, so $S_1 = \sum_j (-1)^j t_j$ $(t_j > 0)$ is alternating, and its value tracks a cosine of the fast phase $w = \sqrt{2/(-\ln q)}$:

$$S_1(q) = 1 - \cos\sqrt{\tfrac{2}{-\ln q}} + O(\sqrt{-\ln q}) \qquad (q \to 1^-), \tag{2}$$

and identically for Σ_1. Hence $S_1 - 1$ straddles 0 infinitely often as $q \to 1^-$ (at the phases $w = (2n+1)\pi$ it is $\approx +1$, at $w = 2n\pi$ it is ≈ -1), so $1 - S_1$ has infinitely many zeros in $(0, 1)$

accumulating at $q=1$ — each a genuine pole of $S_0/(1-S_1)$, since $S_0 \neq 0$ there (its zeros interlace those of $1-S_1$). An algebraic, or even any D-finite, function has only finitely many singularities (they lie among the roots of the leading ODE coefficient); a function with infinitely many poles is therefore neither — hence transcendental. The substitution $q=x^2$ is algebraic, so transcendence passes to $\mathbb{Q}(x)$, and the finite rational block-assembly cannot cancel the infinitely many travel-block poles. $\square$

The rigorous engine is robust: the argument-principle count of poles inside $|q| \le R$ is $0,1,5,46,92$ for $R=0.50,0.65,0.80,0.93,0.95$ (integer-valued, immune to cancellation), manifestly diverging; and an exact search over 421 coefficients finds no algebraic equation ($\deg_F \le 12$, $\deg_x \le 14$) and no D-finite recurrence (order ≤ 16, degree ≤ 20). The asymptotic (2) is moreover *reduced to a single explicit oscillatory-average bound*, no longer a black box. Writing $\hat{t}_j = (2/\tau)^{j+1}/(2j+2)!$ for the leading model term (so $\sum_j (-1)^j \hat{t}_j = 1-\cos w$ exactly), one has the *monotone domination* $0 < t_j \le \hat{t}_j$ with $\rho_j := t_j/\hat{t}_j \downarrow 0$, and Abel summation gives the exact identity

$$S_1(q) = (1-\cos w) + \sum_{j\ge 0} \mu_j\, R_j(w), \qquad \mu_j = \rho_{j-1} - \rho_j \ge 0,\ \textstyle\sum_j \mu_j = 1,$$

with $R_j(w) = \cos w - \sum_{i\le j} (-1)^i w^{2i}/(2i)!$ the cosine Taylor remainder (verified to 110 digits). Lemma 2 is thus exactly the bound $\sum_j \mu_j R_j(w) = O(\sqrt{\tau})$ uniformly in the phase, and its leading term is obtained in *closed form*: the uniform expansion $\log \rho_j = -(j+1)\tau - \frac{1}{9}\tau^2 j^3 + \cdots$ (linear term from the end-factor $g(y) = y/(e^y-1)$, cubic from the cumulative $h(y)^2 = (y/2)^2/\sinh^2(y/2)$) resummed by $\sum_n (-1)^{n-1} n^m \frac{w^{2n}}{(2n)!} = \vartheta^m (1-\cos w)$, $\vartheta = \frac{1}{2} w \partial_w$, gives

$$S_1(q) = 1 - \cos w - \frac{17\sqrt{2}}{36}\sqrt{\tau}\,\sin w + O(\tau),$$

with $-17\sqrt{2}/36 = -0.6678230711\ldots$ confirmed to 15 digits by Richardson extrapolation. (This amplitude is block-specific: $-17\sqrt{2}/36$ is the *bulk* value S_1; the dominant *travel* block Σ_1 has $(\Sigma_1 - 1)/\sqrt{\tau} \to +\sqrt{2}/36$. Both blocks oscillate as $1-\cos w + O(\sqrt{\tau})$ and straddle 1, which is all the theorem uses.) Half of the bulk constant is moreover *unconditional and uniform*: the geometric-collapse split $E := S_1 - (1-\cos w) = T_1 + T_2$ with $T_1 = \cos w - \cos(we^{-\tau/2})$ (exact) obeys $|T_1| \le w(1-e^{-\tau/2}) \le \sqrt{\tau/2}$ for all τ, giving $T_1 = -\frac{1}{\sqrt{2}}\sqrt{\tau}\,\sin w + O(\tau)$ with no hypothesis, and isolates the entire gap in the structurally-smaller residue T_2 (weights $O(\tau^2 j^2)$). Equivalently, via the remainder integral $R_j = \frac{(-1)^{j+1}}{(2j)!}\int_0^w (w-s)^{2j} \sin s\, ds$, one gets the exact convolution $E = \int_0^w \sin s\, K'(w-s)\, ds$ (K' entire, even, $K'(0) = \mu_0 \le \tau$), reducing (2) to $\sup_{[0,w]} |K'| \le C\tau$. In that form the bound is a knife-edge (μ_j fails Hankel positivity, so no Bochner bound; the binding lobe approaches the bound with vanishing margin), but *model subtraction dissolves it*. The geometric weights resum in closed form, $K'_{\text{model}}(u) = -(1-e^{-\tau})\cos(ue^{-\tau/2})$ with $|K'_{\text{model}}| \le \tau$, and corr $:= K' - K'_{\text{model}}$ has $\sup|\text{corr}|/\tau \to \frac{1}{6}$ with the knife-edge gone. The exact resummation engine $\sum_j j^p e^{-j\tau} \frac{(-1)^j u^{2j}}{(2j)!} = (-\partial_\tau)^p \cos(ue^{-\tau/2}) =: D_p$ then expands $K' = \sum_m L_m$ into closed-form layers; L_1 is rigorous, with $\sup|L_1| \le \frac{\tau}{6} + \frac{13\sqrt{2}}{36}\tau^{3/2} + \frac{\tau^2}{6}$ (its fatal $W^3 \sin W$ term cancels to $O(\tau^3)$ by the difference structure). The cancellation in the tail is captured *exactly* by a Touchard/Poisson identity — not a uniform constant (indeed $\sup|D_p|/(w/2)^p \sim 2^{-p}\mathsf{B}_p$ grows like Bell numbers): writing $D_p(W) = 2^{-p}\mathrm{Re}[e^{iW} T_p(iW)]$ (T_p the Touchard polynomial), one has $|D_p(W)| \le 2^{-p} T_p(w)$ and $T_p(w) = \mathbb{E}[N^p]$ for $N \sim \text{Poisson}(w)$, so $\sum_p [j^p] Q\, 2^{-p} T_p(w) = \mathbb{E}[Q(N/2)]$. With the proved R-control $0 \le R_j \le \tau^2 C(j)$ (from $0 \le \log \frac{\sinh(y/2)}{y/2} \le \frac{y^2}{24}$ via Mittag–Leffler), the engine bounds $\sum_{m\ge 2} \sup|L_m| \le 2\,\mathbb{E}[\Psi(\bar{R}^{[2]}(N/2))] = O(\tau)$ ($\Psi(x) = e^x - 1 - x$; $\le 0.0135\,\tau$) — but with one essential subtlety: the *full* absolute majorant $\bar{R} = \sum_n |\lambda_n| \tau^{2n} P_n$ diverges for $j > \pi/\tau$, where the Poisson tail has mass, so one must truncate the τ-series at a finite order ($\bar{R}^{[2]}$, a fixed polynomial) and control the remainder by the pointwise bound

$R_j \le \tau^2 C(j)$ and a Chernoff tail. This gives $\sup|\mathrm{corr}| \le 0.20\,\tau$ and $|T_2| = O(\sqrt{\tau})$, i.e. (2) (end-to-end check $\sup|\mathrm{corr}|/\tau = 0.167$ at $\tau = 10^{-4}$). The constituents — engine, Touchard/Poisson identity, R-control, L_0, L_1 — are individually rigorous, and the truncated assembly closes the tail; the estimate is intricate and we flag it for refereeing before declaring V unconditionally transcendental. We therefore state V transcendental modulo (2); on the corrected (truncated) argument this is reduced to verification of an explicit elementary estimate, with no divergent-majorant gap.

The constituent *blocks*, however, are transcendental *unconditionally*. Each of $\Sigma_1, \Sigma_0, S_1, S_0$ lies in $\mathbb{Z}[[q]]$ (the Lambert factors $2q^m/(1-q^m)$ have integer expansions), has radius of convergence exactly 1 (holomorphic on the disk, integer coefficients), and is non-rational (its diagonal coefficients grow super-polynomially — for Σ_1, $|c_{(j+1)^2}| \ge 2^{j+1}$ — and its Hankel determinants are nonzero through order 12, while a rational function with radius 1 would have $|c_n| = O(n^p)$). By the Pólya–Carlson theorem an integer power series of radius 1 is rational or has $|q| = 1$ as a natural boundary; hence each block has the unit circle as a natural boundary, is not D-finite, and is transcendental over $\mathbb{Q}(x)$ — with no appeal to (2). This does *not* transfer to $V = \Sigma_0/(1-\Sigma_1)$: the boundary essential singularities cancel in the ratio (at a root of unity both numerator and denominator blow up while V stays finite), and $\mathrm{rad}(V) = \sqrt{q^*} \neq 1$, so V-transcendence still routes through the pole-accumulation of $1-\Sigma_1$, i.e. through (2).

Conjecture 28 (Transcendental growth)**.** *The true growth series $\sum_d u_d x^d$ of A396406 is transcendental over $\mathbb{Q}(x)$.*

The true series U and the relaxed V differ by the connectivity correction $d_d := v_d - u_d$ ($u_d \le v_d$, equality for $d \le 7$). Contrary to a first guess this defect is *not* polynomially small: on the 41 computed terms it grows exponentially at essentially the *same* rate as V (ratios d_{d+1}/d_d fall through $1.56, \dots, 1.524$ toward $\beta_2 = 1.4916$, with d_d/v_d rising through $0.02, 0.11, 0.16, 0.20$) and admits no holonomic recurrence of order + degree ≤ 9 (exact). So $D(x) = \sum_d d_d x^d$ shares V's dominant singularity at $x = \sqrt{q^*} = 0.6704\dots$, which *closes* the two elementary routes to U — "D holomorphic past V's first pole" and "D simpler (D-finite) than V" are both false. What survives is closure of D-finite functions under addition: $V = U + D$ is not D-finite (Theorem 27, modulo (2)), so U and D are not *both* D-finite — at least one is transcendental. A sharper handle comes from the connectivity-invariance of Proposition 26: since the travel block is unchanged by the single-Eulerian-trail constraint, U inherits V's travel denominator $1-\Sigma_1$ *exactly*, hence the same infinitely many poles $\{q : \Sigma_1(q) = 1\}$ accumulating at $q = 1$. So U is transcendental *iff* its travel-block numerator N_U avoids vanishing at infinitely many of these poles — the connectivity-corrected analogue of the relaxed numerator non-vanishing $\Sigma_0(q_m) \neq 0$ used for V (itself certified at 58 poles, not in closed form). This is strictly harder than the relaxed case in two honest respects: no closed form for N_U has been validated (the naive "+2-per-cycle" weighting, which would give N_U from Σ_0 by a dilation, miscounts u_d from $d = 9$, where the defect is a *distance*-2 *bridge*, not a unit cycle), and positivity cannot settle it (N_U sign-alternates near $q = 1$). A multi-agent derivation with adversarial verification reduced U-transcendence to exactly this condition and no further; it also surfaced an unexploited lever — the *bulk* denominator $1 - \Sigma_1^{\mathrm{bulk}}$ likewise has infinitely many zeros $\{0.6096, 0.9202, 0.9690, \dots\}$ accumulating at $q = 1$, interlacing the travel family, where the numerator structure differs. U is transcendental unless D exactly cancels these infinitely many poles — not expected, but not yet excluded. Mahler's method offers no shortcut: the catalytic dilation $t \mapsto q^2 t$ shifts the *index* ($\Sigma_k(q) = A_k(q) + C_k(q)\Sigma_{k+2}(q)$ at fixed q), not the base, so neither U, V, nor the blocks is a Mahler function (the base substitution $q \mapsto q^d$ never occurs); equivalently the power-law pole density $\Theta(\varepsilon^{-1/2})$ from $1 - q_n \sim (2/\pi^2)/n^2$ is incompatible with the $O(\log)$ pole count of any d-Mahler function. This realizes, for our tour-cost generating set, the irrational regime of Parry's analysis of wreath-product growth (W. Parry, *Growth series of some wreath products*, Trans. Amer. Math. Soc. 331 (1992), 751–759); the distinguishing feature is the max-coupling dilation, which places A396406 beyond even the algebraic case, in the q-difference (transcendental) regime. The evidence is mutually reinforcing: the exhaustive negatives of Proposition 24 (no rational, low-order D-finite, or low-degree algebraic relation), the dilation structure of the functional equation, and the period-4, $\sim 1/n$ oscillatory convergence of the ratios u_{d+1}/u_d — the hallmark of a q-difference rather than an algebraic dominant singularity, which is why no accelerator could pin β_2 from the terms alone, whereas the closed-form singularity (Proposition 26) gives it exactly.

(c) The growth question becomes a lamplighter growth question. The sequence u_d of A396406 is the growth series of a virtually-$\mathbb{Z} \wr \mathbb{Z}$ group with respect to the geometric *reflection* generating set. The standard lamplighter set gives a rational series (rate φ), whereas the tour-cost set realized by our metric yields a q-difference functional equation; for the *relaxed* growth series this is a transcendental series (Theorem 27, modulo the cosine asymptotic (2)), and transcendence of the true series is conjectural up to the connectivity correction (Conjecture 28). The exact value of β_2 and the relationship between U and V beyond their shared exponential rate remain open problems about lamplighter growth in a non-standard generating set — and the failure of finite presentability (Theorem 21(ii)) explains in hindsight why the Knuth–Bendix completion experiments of the companion document could never terminate. By Theorem 22 the problem is now fully explicit: u_d counts walk-and-deposit configurations of cost d in the edge lamplighter, where a walk on $\mathbb{Z}$ pays one letter per move, bounces for free, and pays one letter per straight-through step or bundled sign change (two if unbundled); the growth series is the counting series of this decorated travelling-salesman model, the same type of computation carried out for other wreath-product metrics by Parry and by Cleary–Taback, but here with lamp group $\mathbb{Z}$ and the reflection cost table. We leave its evaluation — and with it the algebraicity of β_2 — as the now sharply-posed open problem.

The structural facts behind Theorem 4(i) factor through three lemmas.

Lemma 29 (Mayer–Vietoris reduction)**.** *Let $A = V_4 = \langle R_0, R_1 \mid R_0^2, R_1^2, (R_0R_1)^2\rangle$, $B = \mathbb{Z}/2 = \langle R_2 \mid R_2^2\rangle$, so $W = A*B$. Let $Q = D_\infty \times \mathbb{Z}/2$, with generators X, X', Y, c subject to $X^2 = (X')^2 = Y^2 = c^2 = 1$, $XX' = c$, and c central. Let $T_\rho = \ker(W \twoheadrightarrow Q)$. Then, as left $\mathbb{Z}[Q]$-modules,*

$$T_\rho^{\mathrm{ab}} \;\cong\; H_1(W;\mathbb{Z}[Q]) \;\cong\; I_A \cap I_B,$$

and this module is independent of T. Setting $h := (c-1)(Y-1) \in \mathbb{Z}[Q]$, the cyclic submodule $\mathbb{Z}[Q]\cdot h \subseteq I_A \cap I_B$ satisfies $\mathbb{Z}[Q]\cdot h \cong \mathbb{Z}[Q]/(Y+1,\, c+1) =: M$, since the annihilator of h is generated by $(Y+1)$ and $(c+1)$: $(Y+1)\,h = 0$ and $(c+1)\,h = 0$ in $\mathbb{Z}[Q]$. Moreover the equality $I_A \cap I_B = \mathbb{Z}[Q]\cdot h$ holds*, so $T_\rho^{\mathrm{ab}} \cong M$ unconditionally: the proof derives it from the exactness of the chain complex of the Bass–Serre tree of $D_\infty = \langle X\rangle * \langle Y\rangle$.*

Proof. Since $A \hookrightarrow Q$ and $B \hookrightarrow Q$ are injective, $\mathbb{Z}[Q]$ is free as $\mathbb{Z}[A]$- and $\mathbb{Z}[B]$-module, so $H_n(A;\mathbb{Z}[Q]) = H_n(B;\mathbb{Z}[Q]) = 0$ for $n \ge 1$. The Mayer–Vietoris sequence for $W = A * B$ with coefficients $\mathbb{Z}[Q]$ collapses to

$$0 \to H_1(W;\mathbb{Z}[Q]) \to \mathbb{Z}[Q]/I_A \oplus \mathbb{Z}[Q]/I_B \to \mathbb{Z}[Q]/I_W \to 0,$$

and Crowell's exact sequence identifies $T_\rho^{\mathrm{ab}} \cong H_1(W;\mathbb{Z}[Q]) \cong I_A \cap I_B$ as a left $\mathbb{Z}[Q]$-submodule.

Membership. $c = XX' \in A$ gives $c - 1 \in I_A$; centrality of c yields $h = (c-1)(Y-1) = (Y-1)(c-1) \in \mathbb{Z}[Q]\cdot(c-1) \subseteq I_A$. Likewise $Y - 1 \in I_B$ and $h \in \mathbb{Z}[Q]\cdot(Y-1) \subseteq I_B$, so $h \in I_A \cap I_B$.

Annihilator. $(Y+1)(Y-1) = Y^2 - 1 = 0$ and c central, so $(Y+1)h = (c-1)(Y+1)(Y-1) = 0$. Similarly $(c+1)(c-1) = 0$ gives $(c+1)h = 0$.

The equality $I_A \cap I_B = \mathbb{Z}[Q]\cdot h$. Membership and the annihilator computation give $\mathbb{Z}[Q]\cdot h \cong M$ and $\mathbb{Z}[Q]\cdot h \subseteq I_A \cap I_B$; it remains to prove the reverse inclusion. Write $R = \mathbb{Z}[Q]$. Since $X' = cX$, the identity $X' - 1 = c(X-1) + (c-1)$ gives $I_A = R(X-1) + R(c-1)$, while $I_B = R(Y-1)$.

Step 1 (reduction modulo the central ideal). Let $\xi = x(Y-1) \in I_A \cap I_B$ and reduce modulo the two-sided ideal $R(c-1)$ (c is central); the quotient is the group ring $S = \mathbb{Z}[D_\infty]$ of the free product $D_\infty = \langle X\rangle * \langle Y\rangle$. The image satisfies $\bar{x}(Y-1) \in S(X-1) \cap S(Y-1)$. Now the augmented chain complex of the Bass–Serre tree of this free product (vertex set $D_\infty/\langle X\rangle \sqcup D_\infty/\langle Y\rangle$, edge set D_∞) reads

$$0 \to S \xrightarrow{\partial} S/S(X-1) \oplus S/S(Y-1) \to \mathbb{Z} \to 0, \qquad \partial(s) = \big(s \bmod S(X-1),\, -s \bmod S(Y-1)\big),$$

and it is exact because the tree is a tree; exactness at the left — H_1 of a tree vanishes — says precisely $S(X-1) \cap S(Y-1) = 0$. Hence $\bar{x}(Y-1) = 0$ in S, i.e. $x(Y-1) \in R(c-1)$.

Step 2 (annihilator lemma). In any group ring, for an involution Y: $x(Y-1) = 0$ if and only if $x \in R(Y+1)$. (Compare coefficients along the right cosets $\{g, gY\}$: the condition forces $x_g = x_{gY}$ for all g.)

Step 3 (descent). By Step 1, $x(Y-1) = w(c-1)$ for some $w \in R$. Decompose along $R = \mathbb{Z}[D_\infty] \oplus \mathbb{Z}[D_\infty]\,c$: writing $x = x_0 + x_1 c$ and $w = w_0 + w_1 c$, we have $w(c-1) = (w_1 - w_0)(1-c)$, and matching the two components of $x(Y-1) = x_0(Y-1) + x_1(Y-1)\,c$ yields $x_0(Y-1) = -\,x_1(Y-1)$, i.e. $(x_0 + x_1)(Y-1) = 0$. By Step 2, $x_0 + x_1 = z(Y+1)$ for some z. Then $x = x_0(1-c) + (x_0+x_1)c = x_0(1-c) + z(Y+1)c$, and since $(Y+1)(Y-1) = 0$,

$$\xi \;=\; x(Y-1) \;=\; x_0(1-c)(Y-1) \;=\; -\,x_0\,h \;\in\; \mathbb{Z}[Q]\cdot h.$$

No torsion is encountered at any step. □ □

Remark 30 (Finite-quotient cross-check). In each finite quotient $D_{2N} \times \langle c\rangle$ ($t^N = 1$, $3 \le N \le 16$) a direct group-algebra computation gives $\dim(I_A \cap I_B) = \dim(\mathbb{Z}[Q]\cdot h) + 1$, the excess spanned by the rotation-norm element $\nu_N = \sum_{k=0}^{N-1} t^k$ (`code/ideal/intersection_norm.sage`). This is consistent with — and now explained by — the theorem: ν_N is an artifact of the torsion quotient with no analogue in $\mathbb{Z}[Q]$, where $\sum_k t^k$ is not an element of the group ring.

Corollary 31 (The generic kernel is the relative commutator subgroup)**.** *The kernel of the generic side-reflection representation is exactly*

$$\ker \rho_{\text{gen}} \;=\; [T_\rho, T_\rho], \qquad \textit{i.e.} \qquad W_{\text{gen}} \;\cong\; W \,/\, [\ker(W \twoheadrightarrow Q), \ker(W \twoheadrightarrow Q)] :$$

*the generic right-triangle group is the Q-relative metabelianization of $W = V_4 * \mathbb{Z}/2$. In particular the eight length-*10 *relations of Theorem 2 are simply the shortest elements of $[T_\rho, T_\rho]$, and the Mayer–Vietoris module theory of this section and the lamp model of §4.2 describe one and the same object.*

Proof. Linear parts are faithful on Q in the symbolic group (t is a formal variable), so $\ker \rho_{\text{gen}} \subseteq T_\rho$. The image $\rho_{\text{gen}}(T_\rho)$ is the group of pure translations $\mathcal{T}$ (elements with trivial linear part), which is abelian, so $\rho_{\text{gen}}|_{T_\rho}$ factors through T_ρ^{ab} and induces a surjective $\mathbb{Z}[t^{\pm 1}]$-equivariant map $\Phi\colon T_\rho^{\text{ab}} \twoheadrightarrow \mathcal{T}$. By Lemma 29 (now unconditional) $T_\rho^{\text{ab}} \cong M \cong \mathbb{Z}[t^{\pm1}]$, and by Theorem 13 $\mathcal{T} = 2(t-1)\mathbb{Z}[t^{\pm1}] \cong \mathbb{Z}[t^{\pm1}]$. A surjective module map between modules each isomorphic to $\mathbb{Z}[t^{\pm1}]$ is injective (a surjective endomorphism of a Noetherian module is injective), so Φ is an isomorphism and $\ker(\rho_{\text{gen}}|_{T_\rho}) = \ker(T_\rho \to T_\rho^{\text{ab}}) = [T_\rho, T_\rho]$. Concretely, the Crowell generator h realizes as the glide-reflection square: the commutator $[R_0R_1, R_2] = (R_0R_1R_2)^2 = \tau$ (both factors are involutions), verified symbolically, whose translation $2(t^{-1}-1)$ generates $\mathcal{T}$. □

Remark 32 (Universality needs only T-independence, not the explicit M). Theorem 4 requires only that the translation module $T_{\rho_T}^{\text{ab}} \cong I_A \cap I_B$ be *independent of* T. This follows from the Mayer–Vietoris identification, whose right-hand side $I_A \cap I_B$ is a purely group-theoretic object attached to $W = A * B \twoheadrightarrow Q$, with no reference to the geometric realization of any triangle. The cyclic submodule $\mathbb{Z}[Q]\cdot h \cong M = \mathbb{Z}[Q]/(Y+1, c+1)$ established above is used only to write the abstract realization ρ_{abs} down concretely (Lemma 35), which needs only the annihilator relations $(Y+1)h = (c+1)h = 0$; neither the universality conclusion nor ρ_{abs} depends on the conjectural full equality $I_A \cap I_B = \mathbb{Z}[Q] \cdot h$.

Remark 33 (Role of centrality). The cyclic identification works precisely because $c \in Q$ is central. The naive candidate $(X-1)(Y-1)$ fails because X and Y do not commute in D_∞. The central element c supplied by the second $\mathbb{Z}/2$ factor of Q unlocks the cyclic presentation, and is intrinsic to the right-triangle structure: the central involution $c = XX'$ realises the half-turn R_0R_1 around the right-angle vertex.

Lemma 34 (Dilation invariance)**.** *For $\lambda > 0$, $\rho_{\lambda T}$ is conjugate to ρ_T by the central dilation D_λ. In particular $\ker(\rho_{\lambda T}) = \ker(\rho_T)$, so the kernel depends only on the similarity class of T, i.e. only on the shape ratio b/a.*

Proof. $D_\lambda \circ \rho_T(R_i) \circ D_\lambda^{-1} = \rho_{\lambda T}(R_i)$ for $i = 0, 1, 2$, since reflecting across a line and then rescaling about the centre is the same as rescaling and reflecting across the scaled line. Conjugation is a group automorphism of $\mathrm{Aff}(\mathbb{R}^2)$, so $\ker(\rho_{\lambda T}) = \ker(\rho_T)$. □ □

Lemma 35 (Abstract realization)**.** *Define $\rho_{\mathrm{abs}} : W \to Q \ltimes M$ by $R_0 \mapsto (X, 0)$, $R_1 \mapsto (X', 0)$, $R_2 \mapsto (Y, h)$. Then ρ_{abs} is a well-defined group homomorphism.*

Proof. $\rho_{\mathrm{abs}}(R_0)^2 = (X^2, 0) = (1, 0)$ and similarly for R_1; $\rho_{\mathrm{abs}}(R_2)^2 = (Y^2, (Y+1)h) = (1, 0)$ by Lemma 29; $((X, 0)(X', 0))^2 = (c^2, 0) = (1, 0)$. □ □

Proof of Theorem 4. (i) Every ρ_T has the same linear quotient $W \twoheadrightarrow Q$, determined by the reflection pattern and not by the leg lengths: the two legs meet at a right angle, so R_0, R_1 map to the two order-2 generators X, X' of the dihedral/central structure of Q, and R_2 maps to Y, uniformly in T. Hence the translation kernel $T_{\rho_T} = \ker(W \to Q)$ is the same subgroup of W for all T, and by Lemma 29 its abelianization is $T_{\rho_T}^{\mathrm{ab}} \cong I_A \cap I_B$, a purely group-theoretic module independent of T (Remark 32).

(ii) The eight identities of Table 1 hold as 3×3 affine matrices over $\mathbb{Q}(a, b)$ (Theorem 2), so each is in $\ker(\rho_T)$ for every T; combined with the abstract Coxeter relations and the depth-10 completeness of Proposition 3, the collision partition of the length-≤ 10 Coxeter words (and hence u_d^T for $0 \leq d \leq 10$) is the same for every T.

(iii) This is the content of the Lean 4 certificate `universal_layers_through_22` (Remark 8), which deduplicates the symbolic affine images $\rho_{(a,b)}(w) \in \mathrm{Aff}(\mathbb{Q}(a, b))$ over $\mathbb{Q}(a, b)$ and certifies the layer counts $u_0, \ldots, u_{22}$ simultaneously for every right triangle with positive unequal legs.

What this does not prove. By Lemma 35 $\rho_{\mathrm{abs}} : W \to Q \ltimes M$ is well-defined, and for each T the realization ρ_T factors as $\rho_T = (\mathrm{id}_Q \ltimes \Phi_T) \circ \rho_{\mathrm{abs}}$, where $\Phi_T : M \to \mathbb{R}^2$ is the $\mathbb{Z}[Q]$-linear map sending the generator h to the translation $t(\rho_T(R_2))$. This factorization gives the inclusion $\ker(\rho_{\mathrm{abs}}) \subseteq \ker(\rho_T)$ for every T for free. The reverse inclusion $\ker(\rho_T) \subseteq \ker(\rho_{\mathrm{abs}})$, equivalently injectivity of Φ_T, would yield $\ker(\rho_T) = \ker(\rho_{\mathrm{abs}})$ for all T and hence all-depths T-independence. This is in fact *false*: Φ_T is *never* injective for rational (or algebraic) shapes. $M = \mathbb{Z}[Q]/(Y+1, c+1)$ is free of rank 1 over $\mathbb{Z}[t^{\pm 1}]$ $(t = XY)$, hence of infinite rank as an abelian group, while its image under Φ_T lies in the rank-2 abelian group $\mathbb{R}^2$ — and Theorem 15 exhibits an explicit nonzero kernel element, the multiple $2(t-1)\mu_T$ of the shape's minimal polynomial. The all-depths statement of Conjecture 5 fails accordingly (see §4.2). □ □

5. Asymptotic ratio (table of values)

Exact-rational BFS through depth 42 on the reference $(3, 4, 5)$ triangle yields the values in Table 2. The asymptotic behaviour of u_d and the algebraic nature of its generating function are the principal open questions of the program; partial-progress results on conditional transcendence,

p-DFAO state-space analysis, and structural recurrence exclusions appear in the companion document `paper_extra.tex`.

d	u_d	ratio	d	u_d	ratio
0	1	—	22	41 067	1.515
1	3	—	23	62 263	1.516
2	5	1.667	24	94 622	1.520
3	8	1.600	25	143 881	1.521
4	13	1.625	26	217 101	1.509
5	21	1.615	27	327 832	1.510
6	34	1.619	28	495 443	1.511
7	55	1.618	29	749 195	1.512
8	89	1.618	30	1 127 236	1.505
9	144	1.618	31	1 697 179	1.506
10	**225**	1.563	32	2 554 961	1.505
11	351	1.560	33	3 848 384	1.506
12	554	1.578	34	5 777 651	1.501
13	875	1.580	35	8 679 441	1.502
14	1345	1.537	36	13 031 206	1.502
15	2066	1.536	37	19 574 659	1.502
16	3203	1.550	38	29 338 781	1.499
17	4971	1.552	39	43 997 388	1.500
18	7574	1.524	40	65 932 461	1.499
19	11 543	1.524	41	98 849 591	1.499
20	17 683	1.532	42	147 969 934	1.497
21	27 108	1.533			

Table 2: The universal sequence u_d through depth 42.

6. n-dimensional extension: upper bound, closed-form rate, and unconditional Class C faithfulness

The right-triangle analogue in $\mathbb{R}^n$ is the *right-corner orthoscheme*: a simplex with n mutually perpendicular edges meeting at a single vertex. The reflection group W_n is generated by the $n+1$ face reflections.

6.1 The Coxeter group is independent of legs

Lemma 36 (Perpendicular face pairs, verified symbolically)**.** *For the n-dim right-corner orthoscheme with vertices on the staircase $V_0 = 0$, $V_i = V_{i-1} + a_i e_i$, the perpendicular face pairs are exactly $\{(i,j) : |i-j| \geq 2\}$, independent of the positive leg values $(a_1, \dots, a_n)$. In 3D the dihedral cosines are* $\cos\angle(F_0, F_1) = b/\sqrt{a^2+b^2}$, $\cos\angle(F_1, F_2) = ac/\sqrt{(a^2+b^2)(b^2+c^2)}$, $\cos\angle(F_2, F_3) = b/\sqrt{b^2+c^2}$.

The abstract Coxeter group is therefore the right-angled Coxeter group with edges (i,j) for $|i-j| \geq 2$ and $m_{ij} = \infty$ for $|i-j| = 1$, the *same group* for every orthoscheme.

6.2 Steinberg upper bound and closed-form r_n

The commutation graph G_n of the n-dim orthoscheme has vertex set $\{0, \dots, n\}$ and edges (i,j) when $|i-j| \geq 2$, with k-clique count $\binom{n-k+2}{k}$. In 3D, Steinberg's formula gives

$$W(t) \;=\; \frac{(1+t)^2}{1-2t},$$

with $[t^d]W(t) = 9 \cdot 2^{d-2}$ for $d \geq 2$, $u_0 = 1$, $u_1 = 4$, $u_2 = 9$. Since $u_d \leq [t^d]W(t)$ unconditionally:
Theorem 37 (Unconditional 3D upper bound)**.** *For every right-corner orthoscheme in $\mathbb{R}^3$ with positive legs (a, b, c), $u_d \leq 9 \cdot 2^{d-2}$ for $d \geq 2$.*

The Coxeter diagram. For pairwise-distinct legs the Coxeter diagram of W_n is a *path* on $n+1$ nodes (one node per face reflection): by Lemma 36 every non-adjacent face pair $|i-j| \geq 2$ is perpendicular (edge label 2, i.e. no edge drawn), while by Theorem 42 each of the n *adjacent* pairs $|i-j| = 1$ has a dihedral angle that is an irrational multiple of π, hence generates an infinite dihedral group (edge label ∞). Thus the diagram is the path on $n+1$ nodes with all n edges labelled ∞. We do not invoke a closed-form growth rate for this diagram from the literature; instead the rate is computed directly from Steinberg's formula in the proof of Theorem 38 below, and equals the algebraic integer stated there (e.g. $r_2 = \varphi$, $r_3 = 2$, consistent with the $n = 3$ series $(1+t)^2/(1-2t)$).
Theorem 38 (Closed-form n-dim Coxeter rate)**.** *For every $n \geq 2$, the abstract orbit-growth rate of the n-dim right-corner orthoscheme reflection group is*

$$r_n \;=\; 1 \;+\; 2\cos\!\left(\frac{2\pi}{n+3}\right).$$

Proof. By Steinberg's formula applied to G_n, $1/W_n(t^{-1}) = \sum_k (-1)^k \binom{n-k+2}{k}(1+t)^{-k}$. Substituting $y = -1/(1+t)$ collapses the sum to the Fibonacci–Pell polynomial $P_{n+3}(y) = \sum_k \binom{n-k+2}{k} y^k$, defined by $P_0 = 0$, $P_1 = 1$, $P_m = P_{m-1} + yP_{m-2}$. The characteristic equation $x^2 = x + y$ has roots whose ratio $(a/b)^m = 1$ gives $y_j = -1/(4\cos^2(\pi j/m))$ for $j = 1, \ldots, \lfloor (m-1)/2 \rfloor$. Translating back via $y = -1/(1+t)$ and using $4\cos^2\theta - 1 = 1 + 2\cos(2\theta)$:

$$t_j \;=\; 1 + 2\cos\!\left(\frac{2\pi j}{n+3}\right);$$

the asymptotic rate is the maximum, attained at $j = 1$. □ □

Corollary 39. *$3 - r_n = 4\sin^2(\pi/(n+3)) \sim 4\pi^2/(n+3)^2$, so $r_n \to 3$.*
Remark 40 (Dynamical-degrees connection). The family $\{1 + 2\cos(2\pi/m)\}_{m \geq 3}$ also arises in birational surface dynamics (McMullen 2007; Bedford–Kim 2009). For $n \neq 3$, r_n is Pisot or totally-real of degree ≥ 3, hence *not* a Salem number, and by McMullen 2007 Theorem 1.5 cannot be the dynamical degree of a projective rational-surface automorphism.

6.3 Class C unconditional faithfulness

Lemma 41 (Uniform Schur-complement determinant identity)**.** *For every $n \geq 3$ and every positive tuple $(a_1, \ldots, a_n)$, the $(n+1) \times (n+1)$ Lorentzian Gram matrix Q_n of the n-dim right-corner orthoscheme satisfies*

$$\det Q_n \;=\; -\prod_{i=1}^{n} a_i^2.$$

Here Q_n is the symmetric tridiagonal matrix with $Q_n[0,0] = 1 - a_1^2$, $Q_n[i,i] = a_i^2 + a_{i+1}^2$ ($1 \leq i \leq n-1$), $Q_n[n,n] = 1$, $Q_n[0,1] = a_2$, $Q_n[i,i+1] = a_i a_{i+2}$ ($1 \leq i \leq n-2$), $Q_n[n-1,n] = a_{n-1}$, read from the staircase normal vectors with the Lorentzian endpoint correction.

Proof. Let $D_k = \det Q_n[:k{+}1, :k{+}1]$. Tridiagonal expansion gives $D_k = Q_n[k,k]D_{k-1} - Q_n[k-1,k]^2 D_{k-2}$ with $D_{-1} = 1$. We claim $D_k = P_k := (\prod_{i=1}^{k} a_i^2)(1 - \sum_{i=1}^{k+1} a_i^2)$ for $1 \leq k \leq n-1$, and $D_n = -\prod_{i=1}^{n} a_i^2$.

Base. $D_0 = 1 - a_1^2$; $D_1 = (1 - a_1^2)(a_1^2 + a_2^2) - a_2^2 = a_1^2(1 - a_1^2 - a_2^2) = P_1$.

Interior step. For $2 \le k \le n-1$, the recurrence reduces to the polynomial identity (in the symbols $a_{k-1}, a_k, a_{k+1}, S_{k-1}, P_{k-2}$ with $S_{k-1} = \sum_{i=1}^{k-1} a_i^2$, $P_{k-2} = \prod_{i=1}^{k-2} a_i^2$)

$$P_{k-2}\, a_{k-1}^2 a_k^2\, (1 - S_{k-1} - a_k^2 - a_{k+1}^2) = (a_k^2 + a_{k+1}^2)\, P_{k-2}\, a_{k-1}^2\, (1 - S_{k-1} - a_k^2) - (a_{k-1}a_{k+1})^2\, P_{k-2}\, (1 - S_{k-1}),$$

which expands to zero in $\mathbb{Z}[a_{k-1}, a_k, a_{k+1}, S_{k-1}, P_{k-2}]$ and is verified symbolically. The identity is independent of k and of n. The general-n determinant identity is thus a complete mathematical proof by the standard tridiagonal cofactor (three-term) expansion together with this induction. The individual *algebraic steps* of the induction (base, $k = 2$, inductive, and boundary) are additionally ring-checked in Lean 4 (theorems `schur_base_step`, `schur_k2_step`, `schur_inductive_step`, `schur_boundary_step` in `lean/with_mathlib/SchurGeneral.lean`, each discharged by the `ring` tactic and hence kernel-checked); Lean does *not* prove the general-n determinant identity, which is the mathematical induction above. Separately, the determinant identity $\det Q_n = -\prod_{i=1}^n a_i^2$ is checked by `native_decide` for $n = 2, \ldots, 10$ on explicit leg sequences in `lean/RightTriangleReflection.lean`.

Final step. At $k = n$, $D_n = D_{n-1} - a_{n-1}^2 D_{n-2}$, and

$$D_{n-1} - a_{n-1}^2 D_{n-2} = P_{n-2}\, a_{n-1}^2 \big[(1 - S_{n-1} - a_n^2) - (1 - S_{n-1})\big] = -\prod_{i=1}^{n} a_i^2.$$

The three identities (base, interior, final) close the induction for every $n \ge 3$. □ □

Theorem 42 (Class C unconditional faithfulness, all $n \ge 3$)**.** *For every $n \ge 3$ and every leg tuple $(a_1, \ldots, a_n) \in \mathbb{Q}_{>0}^n$ with pairwise distinct coordinates, the affine representation $\rho : W_n \to \mathrm{Aff}(\mathbb{R}^n)$ of the n-dim right-corner orthoscheme reflection group is faithful, and*

$$u_d(O) \;=\; [t^d]\, W_n(t) \qquad \textit{for every } d \ge 0.$$

In particular for $n = 3$ this is $u_d = 9 \cdot 2^{d-2}$ for $d \ge 2$.

Proof. Write $S = \{R_0, \ldots, R_n\}$ for the face reflections and let $m_{ij} \in \{2, 3, \ldots, \infty\}$ be the order of $R_i R_j$ in the group $W_n^{\mathrm{geo}} = \langle S \rangle$ actually generated inside $\mathrm{Isom}(\mathbb{R}^n)$. Faithfulness of ρ together with the identity $u_d(O) = [t^d]\, W_n(t)$ is equivalent to the assertion that (m_{ij}) equals the intended right-angled matrix

$$m_{ij} = 2 \ (|i-j| \ge 2), \qquad m_{i,i+1} = \infty \ (0 \le i \le n-1), \tag{$*$}$$

i.e. that the orthoscheme introduces *no* relation $(R_i R_j)^m = 1$ beyond the perpendicular pairs. We prove ($*$); the theorem then follows from Tits' theorem applied to the resulting Coxeter system (end of proof). Non-degeneracy of Q_n (Lemma 41) does *not* by itself give faithfulness: a non-degenerate Gram form is compatible with a rational dihedral angle. The isosceles case $a = b$ has $\det Q_n = -\prod a_i^2 \ne 0$ yet $\theta_{01} = \pi/4$, giving the extra relation $(R_0 R_1)^4 = 1$, the cd $= 4$ case of Theorem 43. The argument therefore reduces to showing every non-perpendicular dihedral angle is an irrational multiple of π; $\det Q_n \ne 0$ enters only to confirm the simplex is non-degenerate for all positive legs.

Step 1 (dihedral angles $\leftrightarrow$ relations). For two faces meeting along a codimension-2 ridge, R_i, R_j generate a finite dihedral group of order $2m$ iff the dihedral angle $\theta_{ij} = \pi/m$ for an integer $m \ge 2$, and an infinite dihedral group (no relation) iff $\theta_{ij} \notin \pi\mathbb{Q}$. By Lemma 36 the pairs with $|i-j| \ge 2$ are perpendicular ($m_{ij} = 2$), giving the right-angled edges of ($*$). It remains to show every non-perpendicular $\theta_{i,i+1}$ is an irrational multiple of π.

Step 2 (Niven–Conway–Jones constraint). By Lemma 36 each non-perpendicular $\cos^2\theta_{i,i+1}$ is rational in the legs. If some $\theta_{i,i+1} \in \pi\mathbb{Q}$, then by the extended Niven theorem (Conway–Jones 1976) a rational angle with rational squared cosine satisfies $\cos\theta_{i,i+1} \in \{0, \pm\frac{1}{2}, \pm\frac{\sqrt{2}}{2}, \pm\frac{\sqrt{3}}{2}, \pm 1\}$, i.e. $\cos^2\theta_{i,i+1} \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$. The values 0 (perpendicular) and 1 (degenerate simplex) are excluded, so an extra relation requires $\cos^2\theta_{i,i+1} \in \{\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$, i.e. $\theta_{i,i+1} \in \{\pi/3, \pi/4, \pi/6\}$.

Step 3 (boundary dihedrals; elementary). The end edges have $\cos^2\theta_{01} = b^2/(a^2+b^2)$ in the local two-leg notation $(a,b) = (a_1, a_2)$, and symmetrically at the far end. Setting this equal to $\frac{1}{4}, \frac{1}{2}, \frac{3}{4}$ forces $b^2 = \frac{1}{3}a^2$, $b^2 = a^2$, $b^2 = 3a^2$ respectively: the first and third give $b/a \in \{1/\sqrt{3}, \sqrt{3}\} \notin \mathbb{Q}$, contradicting rationality of the legs; the middle gives $a = b$, excluded by the pairwise-distinct hypothesis. No boundary dihedral is a rational multiple of π, and no elliptic curve is needed here.

Step 4 (interior dihedrals; square-discriminant → Mordell). An interior edge involves three consecutive legs $(a,b,c) = (a_i, a_{i+1}, a_{i+2})$ with b the middle leg, and by Lemma 36

$$\cos^2\theta_{i,i+1} = \frac{a^2c^2}{(a^2+b^2)(b^2+c^2)}.$$

Setting this equal to a Niven value $\frac{1}{4}, \frac{1}{2}, \frac{3}{4}$ and clearing denominators yields a quadratic in $B := b^2$. Since the legs are rational, $B = b^2 \in \mathbb{Q}$, so this quadratic has a rational root only if its discriminant is a rational square. The three discriminants are, up to a positive rational square factor, exactly the Diophantine varieties

$$\mathcal{V}_{1/4}: \ a^4 + 14a^2c^2 + c^4 = m^2, \quad \mathcal{V}_{1/2}: \ a^4 + 6a^2c^2 + c^4 = k^2, \quad \mathcal{V}_{3/4}: \ 9a^4 + 30a^2c^2 + 9c^4 = k^2$$

in the two flanking legs (this square-discriminant condition is the correct origin of the three varieties). A rational interior angle thus forces a rational point on one of $\mathcal{V}_{1/4}, \mathcal{V}_{1/2}, \mathcal{V}_{3/4}$ with $a, c \in \mathbb{Q}_{>0}$. Each is a genus-1 curve in $t = a/c$ whose Jacobian has rank 0, so each has only finitely many rational points:

- $\mathcal{V}_{1/4}$: Jacobian $y^2 = x(x-12)(x-16)$, Cremona `24a1` (conductor 24, rank 0, torsion $\mathbb{Z}/4 \oplus \mathbb{Z}/2$); rational points $t = a/c \in \{-1, 0, 1\}$.
- $\mathcal{V}_{1/2}$: Jacobian $y^2 = x^3 - x$, Cremona `32a2` (conductor 32, rank 0, torsion $\mathbb{Z}/2 \oplus \mathbb{Z}/2$, the four points $(0,0), (\pm 1, 0), \infty$); rational points $t = 0$.
- $\mathcal{V}_{3/4}$: Jacobian $y^2 = x(x-300)(x-1200)$, Cremona `72a2` (conductor 72, rank 0, torsion $\mathbb{Z}/2 \oplus \mathbb{Z}/2$); rational points $t = 0$.

Every such point gives $ac = 0$ or $a = \pm c$, all excluded by the pairwise-distinct, strictly positive hypothesis on the legs. (Ranks, torsion, Cremona labels, and the rational-point determinations are verified in SageMath in `code/mordell/mordell_24a1.sage`, `mordell_32a2.sage`, `mordell_72a2.sage`, with output committed in the project repository.) Hence no interior dihedral is a rational multiple of π either.

Conclusion. By Steps 3–4 no non-perpendicular $\theta_{i,i+1}$ lies in $\pi\mathbb{Q}$, so $m_{i,i+1} = \infty$, and with Step 1 the geometric Coxeter matrix is exactly $(*)$. Thus W_n^{geo} is the Coxeter system with matrix $(*)$, and Tits' theorem (Humphreys, *Reflection Groups and Coxeter Groups*, Thm. 5.4) applied to *this established system* gives faithfulness of the geometric representation ρ. Consequently $u_d(O) = \#\{w : \ell(w) = d\} = [t^d]\, W_n(t)$; for $n = 3$ this is $9 \cdot 2^{d-2}$ for $d \geq 2$. □ □

6.4 The collision-depth invariant $\mathrm{cd}_n \in \{\infty, 4, 3\}$

For a leg sequence $(a_1, \dots, a_n)$ define

$$e := \#\{i \in \{1, n-1\} : a_i = a_{i+1}\} \in \{0,1,2\}, \qquad t := \#\{i : 1 \leq i \leq n-2,\ a_i = a_{i+1} = a_{i+2}\}.$$

e counts *endpoint pairs* (adjacent equal legs at the leftmost or rightmost positions) and t counts *triples*. Let $\mathrm{cd}_n(a_1, \dots, a_n)$ be the minimal word-length at which the orbit BFS first exhibits a non-trivial collision (i.e. the first depth at which $u_d < [t^d]W_n(t)$); set $\mathrm{cd}_n = \infty$ when the affine representation is faithful at every length.

Theorem 43 (Collision-depth invariant, all $n \geq 3$)**.** *For every $n \geq 3$ and every $(a_1, \ldots, a_n) \in \mathbb{Q}^n_{>0}$,*

$$\mathrm{cd}_n(a_1, \ldots, a_n) = \begin{cases} \infty & \text{if } e = t = 0, \\ 4 & \text{if } e \geq 1,\ t = 0, \\ 3 & \text{if } t \geq 1. \end{cases}$$

Proof. Faithfulness for $e = t = 0$ is Theorem 42. An endpoint pair $a_i = a_{i+1}$ at a boundary makes the corresponding boundary cosine $b/\sqrt{a^2+b^2}$ equal $\pm\sqrt{2}/2$, producing the length-4 relation $(R_iR_{i+1})^4 = \mathrm{id}$ and hence cd ≤ 4; no shorter relation can appear because $\cos^2\theta = 0, 1$ are degenerate and the $\pm 1/2$, $\pm\sqrt{3}/2$ cases reduce as in the proof of Theorem 42 to empty or degenerate loci. An interior triple $a_i = a_{i+1} = a_{i+2}$ makes the middle cosine $ac/\sqrt{(a^2+b^2)(b^2+c^2)} = 1/2$ (after the quadratic-in-b^2 analysis with $a = b = c$), producing the length-3 braid relation $R_iR_{i+1}R_i = R_{i+1}R_iR_{i+1}$ and hence cd ≤ 3. An interior pair (with no endpoint pair and no triple) leaves all flanking dihedrals irrational multiples of π, by the following elementary argument. For an isolated interior equal pair $a_i = a_{i+1} = a$ with neighbouring leg c ($c \neq a$, else a triple), the relevant dihedral is governed by three consecutive legs (a, a, c), so by Lemma 36 (with middle leg equal to a)

$$\cos^2\theta = \frac{a^2c^2}{(a^2+a^2)(a^2+c^2)} = \frac{c^2}{2\,(a^2+c^2)},$$

not the generic distinct-leg expression. Setting this equal to the Niven values gives: $\frac{1}{4} \Rightarrow c^2 = a^2$ (i.e. $c = a$, a triple, excluded); $\frac{1}{2} \Rightarrow$ no solution ($c^2 = a^2 + c^2$ is impossible); $\frac{3}{4} \Rightarrow c^2 = -3a^2$, impossible over the reals. The values $0, 1$ are degenerate. Hence no flanking dihedral of an isolated interior pair is a rational multiple of π (an elementary computation requiring no elliptic curve), so cd $= \infty$. (The $\cos^2\theta = c^2/(2(a^2+c^2))$ formula and the three Niven specializations are confirmed symbolically in SageMath.) □ □

In $n = 3$ every adjacent equal pair is automatically an endpoint pair, so the formula reduces to the run-length statistic; in $n \geq 4$ *interior* pairs $a_i = a_{i+1}$ (e.g. $a_2 = a_3$ in $(1, 2, 2, 3)$) leave cd $= \infty$, distinguishing (e, t) from the longest-run statistic.

7. Reproducibility

All computations use exact rational arithmetic; the symbolic verification of Theorem 2 uses SymPy in $\mathbb{Q}(a, b)$, and the Schur-complement identity of Lemma 41 is verified symbolically in $\mathbb{Z}[a_{k-1}, a_k, a_{k+1}, S_{k-1}, P_{k-2}]$. The rank-0 descents on the Cremona curves `24a1`, `32a2`, and `72a2` (for $\mathcal{V}_{1/4}, \mathcal{V}_{1/2}, \mathcal{V}_{3/4}$ respectively) are verified in SageMath via `EllipticCurve.rank()` and a direct rational-point determination, in `code/mordell/`. The BFS values in Table 2 are computed by an exact-rational BFS through depth 25 and a Rust disk-streaming BFS through depth 42. Source code and reproduce scripts accompany this paper in the project repository.

Formal (machine-checked) verification. Several of the computational claims are additionally certified in the Lean 4 proof assistant, in the `lean/` subtree of the project repository. A core, Mathlib-free development (`RightTriangleReflection.lean`) machine-checks the basic Coxeter relations, all eight length-10 relations and their explicit affine matrices on the $(3, 4, 5)$ triangle, a direct breadth-first reconstruction of $a(0), \ldots, a(17)$, the Fibonacci coincidence $a(n) = F_{n+3}$ for $1 \leq n \leq 9$, and the determinant identity $\det Q_n = -\prod a_i^2$ for $n = 2, \ldots, 10$. A second development depending on Mathlib (`with_mathlib/`) proves the eight length-10 relations symbolically over $\mathbb{Q}(a, b)$ (Theorem 2), the algebraic steps of the Schur induction (Lemma 41), and the T-independence of the orbit-growth sequence through depth 22 (Theorem 4(iii), Remark 8), the last holding simultaneously for all right triangles with unequal legs. The all-depths statement (Conjecture 5) is *false* (Theorem 15); the explicit witness words are verified in exact rational arithmetic by `code/zeta_probe/witness.py`, not in Lean. The `ring`-based theorems (the symbolic length-10 relations and the Schur induction steps) are verified entirely by the Lean kernel; the remaining results use `native_decide`, whose trusted base additionally includes the Lean compiler and the correctness of the hand-rolled `partial` functions for polynomial determinant and deduplication. Building the projects re-verifies every claim.

On the use of AI

This paper was produced through extended human–AI collaboration, and we state the division of labour plainly. The author is an independent researcher (and AI red-teamer), not a professional mathematician. The mathematical development was AI-led under human direction: the principal model was Anthropic's Claude, cross-checked against other large language models in a multi-model, adversarial, peer-review-style process that caught and corrected several errors during development. Concretely, the AI did the bulk of the work of deriving and writing the proofs and the text; it wrote the exact-rational breadth-first search and the Rust disk-streaming implementation; it produced the SymPy and SageMath symbolic verifications and the elliptic-curve (rank-0 descent) computations for the three Cremona curves; and it carried out the Lean 4 formalization, including the eight length-10 relations and the Schur induction steps as kernel-checked `ring` proofs and the depth-22 universality certificate via `native_decide`.

The author's role was to direct the investigation, to choose and pose the problems and the lines of attack, to run the adversarial cross-checking between models, and to verify that the computational artifacts (the BFS code, the SymPy/SageMath scripts, and the Lean developments) actually run and reproduce the stated results. The author takes responsibility for the final text and for any remaining errors. The formal Lean 4 certificates are offered precisely so that the central machine-verifiable claims do not rest on trust in either the human or the AI contributors: they are checked by the Lean kernel (or, where noted, by `native_decide`) and can be re-run independently.

References

- J. E. Humphreys, *Reflection Groups and Coxeter Groups*, Cambridge Studies in Advanced Mathematics 29, Cambridge University Press, 1990. Tits' faithfulness theorem (Thm. 5.4) and Steinberg's word-growth formula (§5.12).
- I. M. Niven, *Irrational Numbers*, Carus Mathematical Monographs 11, MAA, 1956.
- J. H. Conway, A. J. Jones, *Trigonometric Diophantine equations (On vanishing sums of roots of unity)*, Acta Arith. 30 (1976), 229–240.
- R. Steinberg, *Endomorphisms of Linear Algebraic Groups*, Memoirs of the AMS 80, 1968.
- L. J. Mordell, *Diophantine Equations*, Pure and Applied Mathematics 30, Academic Press, 1969. §17 (rank-0 descent on $x^4 + 6x^2y^2 + y^4 = z^2$).
- J. W. S. Cassels, *Lectures on Elliptic Curves*, LMS Student Texts 24, Cambridge University Press, 1991. Ch. II on Fermat-style descent.
- J. E. Cremona, *Algorithms for Modular Elliptic Curves*, 2nd edition, Cambridge University Press, 1997; online tables at `https://www.lmfdb.org/EllipticCurve/Q/`. Curves `24a1`, `32a2`, and `72a2`.

- M. W. Davis, *The Geometry and Topology of Coxeter Groups*, LMS Monographs 32, Princeton University Press, 2008.
- J. W. Cannon, P. Wagreich, *Growth functions of surface groups*, Math. Ann. 293 (1992), no. 2, 239–257.
- W. J. Floyd, S. P. Plotnick, *Growth functions on Fuchsian groups and the Euler characteristic*, Invent. Math. 88 (1987), no. 1, 1–29.
- H. S. M. Coxeter, *Discrete groups generated by reflections*, Annals of Math. 35 (1934), 588–621.
- C. T. McMullen, *Dynamics on blowups of the projective plane*, Publ. Math. IHES 105 (2007), 49–89.
- E. Bedford, K. Kim, *Dynamics of rational surface automorphisms: linear fractional recurrences*, J. Geom. Anal. 19 (2009), no. 3, 553–583.
- J. Diller, C. Favre, *Dynamics of bimeromorphic maps of surfaces*, Amer. J. Math. 123 (2001), no. 6, 1135–1169.
- S. Isola, R. Piergallini, *On the generic triangle group*, arXiv:1405.1881 [math.MG], 2014–2015.
- L. Sadun, *Some generalizations of the pinwheel tiling*, Discrete Comput. Geom. 20 (1998), 79–110; arXiv:math/9712263.
- D. Frettlöh, *Substitution tilings with statistical circular symmetry*, European J. Combin. 29 (2008), no. 8, 1881–1893.
- OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, `https://oeis.org`. Sequence A000045 (Fibonacci numbers) and Sequence A396406 (the universal right-triangle reflection sequence u_d).